University of Würzburg
Faculty of Business Management and Economics
Chair of International Economics
Prof. Dr. Michael Pflüger

Master's thesis
"Regional Misallocation: Italy and Germany compared"
Advanced Macroeconomics
Supervisor: Michael Pflüger
Submitted on 2.10.2024 by: Michele Corbisiero

Michele Corbisiero
Matr.-Nr.: 3009589
Friedenstraße 2, 97072, Würzburg
michele.corbisiero@stud-mail.uni-wuerzburg.de

Zusammenfassung der Abschlussarbeit

Diese Analyse untersucht die lokalen und aggregierten Auswirkungen verschiedener Lohnfestsetzungssysteme auf Löhne, Nichtbeschäftigungsquoten, Wohnkosten und Reallöhne und vergleicht die räumliche Verteilung in Italien und Deutschland. Die Lohnungleichheit ist in Europa weit verbreitet und nimmt zu. Diese Studie zeigt, dass politische Maßnahmen, die die Lohnungleichheit abmildern, auch zu kostspieligen Beschäftigungsunterschieden zwischen verschiedenen Makroregionen führen können, was regionale Fehlallokationen provoziert. Die Analyse konzentriert sich auf die Fälle Italien und Deutschland, die für einen Vergleich geeignet sind, weil sie zwei unterschiedliche Lohnfestsetzungssysteme haben, aber dieselbe geografische Produktivitätsverteilung aufweisen. Es wurde ein theoretisches Modell mit zwei verschiedenen Szenarien entwickelt: ein Szenario des freien Marktes und ein Szenario mit festen Löhnen. Im Szenario der freien Marktwirtschaft (ein Extremfall des deutschen Systems) sind die Beschäftigungsquote, das Kapital, der Nominallohn und der Hauspreis in der produktivsten Region höher. Im Szenario mit festen Löhnen (einem Extremfall des italienischen Systems) sind die Beschäftigungsquote, das Kapital und die Immobilienpreise in der produktivsten Region höher, aber die Reallöhne sind in der am wenigsten produktiven Region höher (aufgrund der niedrigen Preise). Darüber hinaus gibt es im Szenario mit festen Löhnen ein großes regionales Gefälle bei der Beschäftigungsquote. Kurz gesagt, die Lohnungleichheit ist im Szenario des freien Marktes ausgeprägter und die Ungleichheit bei der Beschäftigung ist im Szenario der festen Löhne stärker. Eine weitere wichtige Implikation des Modells ist, dass das Arbeitseinkommen im Szenario des freien Marktes höher ist als im Fall der Lohnbindung. In dieser Studie wird festgestellt, dass ein zentralisiertes und starres Lohnfestsetzungssystem die Lohnungleichheit abschwächt, aber die regionale Fehlallokation verstärkt, insbesondere in Ländern mit großen internen Produktivitätsunterschieden, da die Löhne nicht an die Produktivität angepasst werden. Unsere Analyse belegt daher, dass die regionalen Fehlallokationen in Italien auf die Starrheit des Lohnfindungssystems zurückzuführen sind. Das soll nicht heißen, dass Italien die gleichen Maßnahmen ergreifen sollte, die Deutschland seit der Wiedervereinigung ergriffen hat, um allokative Ineffizienz zu verringern.

List of Contents

1. Introduction 3

2. Wage setting system 5

2.1 Italy 5

2.2 Germany 10

2.3 European Union countries 12

3. Theoretical Framework 15

3.1 Configuration 16

3.2 Free-market scenario 17

3.3 Fixed-wage case 19

3.4 Aggregate Effects 22

4. Empirical Evidence 25

4.1 Data description 25

4.2 Productivity (Value Added) 27

4.3 Nominal wages 29

4.4 Employment rate 33

4.5 Price level and Real wages 37

5. Counterfactual analysis 43

5.1 Average hourly wage 46

5.2 Employment rate 47

5.3 Aggregate labour income per capita 48

6. Discussion 50

7. Conclusion 56

References 58

1. Introduction

Wage inequality is widespread and growing in Europe. Countries, through labour market institutions, try to slow down wage inequality by pursuing policies that aim at wage equalisation. This study show that such policies can mitigate wage inequality, but that they can also create costly employment disparities between different macro-regions, provoking regional misallocation. This analysis focus on the cases of Italy and Germany, that are valid for comparison because they have two different wage setting systems, but the share the same geographic productivity distribution, as western Germany and northern Italy are more productive than east and southern Italy[1]. The Italian system is rigid because it does not allow wages to adapt to the local labour market, while the German is flexible, since wages fit with the local market in each region. The Italian system pursues wage equalisation, but this leads to a non-correlation between local productivity and local nominal wages. As a result, firms in the south, while being less productive than firms in the north, face almost the same labour costs. Consequently, firms in the south fail to achieve an optimal level of production, which also leads to high non-employment rate and thus low labour income. This study states that if Italy succeed in relating local productivity to the local nominal wage like Germany, aggregate employment and aggregate labour income would increase by 11.04% and 7.51%. It is important to note that this thesis incorporates the analysis, models, data and calculations of the paper 'Wage equalization and regional misallocation: evidence from Italian and German provinces' by Boeri, Moretti, Ichino and Posch (2021). The research is based on this paper because it is one of the most in-depth and detailed analyses in the literature for this subject.

Labour market policies are one of the causes of wage and employment inequality, as reforms impact employment and productivity establishing a system of rules that defines wage setting system. The grade of flexibility of a wage setting system directly influences the freedom of firms to fix wages at the local market level and, so, production capacity and employability. If firms have more flexibility to fix wages, they can produce and hire more[2] (Gnocato, Modena, Tomasi 2020, p.1).

[1] Geographic productivity distribution is very similar between Italy and Germany, not only qualitatively, but also quantitatively. The difference in productivity between northern and southern Italy is 17.6%, while it is 22.8% for western and eastern Germany (Chapter 4).

[2] A wage structure with little diversification would thus have represented a systematic cause of unemployment, particularly in southern Italy. In this region, nominal wages set above equilibrium levels would have created advantages only for employed workers. Unemployed suffer from "gabbie salariali", wage cages in English (Poy 2017, p.21).

Instead, if firms have a rigid system, they cannot pay employees below a minimum wage, and they could encounter difficulties to produce at optimum level and hire new employees. Italy is an example of this as, due to the rigidity of the system, companies in South cannot lower wages on the market level. Because of the rigid and centralized wage setting mechanism, with same rules for every region, northern and southern firms must fix the same, or similar, minimum wage. North Italy and West Germany are much more productive than South and East respectively. Even though it is common for a country to have internal differences in terms of productivity[3], what makes the similarities between Italy and Germany special is the clear contrast between two macro regions. However, since the two diverse wage setting mechanism, productivity internal split leads to opposite outcomes. In fact, Germany shows internal wage inequality, while Italy presents an important internal employment inequality, causing regional misallocation[4]. The study argues that it is opportune for Italy to make its system more to remedy this inefficiency. It is essential to point out that the causes of the productivity internal differences in Italy and Germany are not the objective of the study and will not be dealt with in the analysis. The study focuses on the existing dynamics between the system and economic variables to understand how a flexibilization of the system would lead an increase on employment and aggregate labour income in southern Italy.

Chapter 2 deepens into Italian and German wage setting and provides and overview of the solutions for wage setting in other European countries. Chapter 3 suggests a theoretical framework for this research that gives us some important implications. Afterwards, Chapter 4 deals with the empirical evidence, where real data on wages, productivity, employment and prices for Italy and Germany are presented. Chapter 5 shows calculations and results counterfactual scenarios, in which Italy adopts wages on productivity as Germany. That is necessary to measure the aggregate costs of spatial misallocation. Chapter 6 adds some insights on the effect of welfare and local amenities on market of labour in the discussion, before the conclusions.

[3] Internal differences in productivity are also normal in other countries, as they often result from the urbanisation process, since people are attracted by the productive cities. This creates heterogeneity in a country in term of productivity. There is a big productivity difference between urbanised and non-urbanised areas. A doubling of the size of a city is correlated with a 5.98% increase in productivity (Sveikauskas 1975).

[4] The main reason for which southern firms are less productive and competitive than northern firms is the more severe spatial inefficiency of southern regions. In these, the market fails to allocate resources efficiently (Erbetta and Petraglia 2011, p.2).

2. Wage setting system

This chapter describes the features of wage setting mechanism in Italy and in Germany and gives a panoramic of other European countries to assess whether it is appropriate for Italy to follow other Mediterranean countries that transform their system in recent years.

In Germany, the decentralization of the wage bargaining system has influenced the growth of wage inequality, while in Italy the centralized system has limited wage dispersion between companies, although it increased employment inequality between the northern and southern regions.

It is necessary to focus on the activities of trade unions and how labour market institutions handle wage negotiations, to understand the overall dynamics of the phenomenon. In fact, the degree of flexibility of the system depends on how the relationship between trade unions and employers is regulated. Therefore, trade union activity has a decisive effect on regional growth and allocative efficiency (Faini 1999, p.16).

Understanding the characteristics and the differences of labour market institutions is important to catch the divergences in wage and employment rate geographic distribution. The following subchapter investigates the history of wage negotiations in Italy to better understand the reasons of a centralized wage system. Although, it is difficult to explore in detail the complexity of the mechanisms, but it is still possible to catalogue them according to their degree of flexibility and decentralisation.

2.1 Italy

Italy remains, along with Denmark, Cyprus, Sweden, Finland and Austria, one of the few EU countries to not have any "universal" minimum wage level[5]. Despite that, Italian firms are not allowed to lower salaries arbitrarily. In fact, national agreements between union and employees, deriving from collective bargaining, impose lots of

[5] The last proposal of law (2023) was rejected by Meloni's government. The opposition parties have submitted the to the Chamber of Deputies, 4th July 2023, a bill for the establishment of a minimum wage. The proposal includes a minimum hourly wage of 9 euros per hour. The bill is one of the possible attempts to implement the provisions of the EU Directive 2022/2041 on the minimum wage. However, the Chamber of Deputies has approved the government's delegation, which effectively halts the minimum wage bill proposed by the opposition. There were 153 votes in favour, 118 against, and 3 abstentions.

sectorial minimum wages. These 346 national agreements cover most of the Italian labour force, specifically the 97,7% of employees are members of the social security system[6] and 99.3% of employers with their companies (Boeri et al. 2021, p. 7). In addition, when trade unions and employers' organizations are not sufficiently representative, administrative extensions are used to broaden collective bargaining and make it more inclusive, even for workers and employers who do not belong to the signatory unions or organizations (OECD 2018, p. 31).

The rigidity of the Italian wage setting mechanism comes from the centralization of national contract. In Italy, where the same wage policies are applied across the entire country, the economic development is harmed by the more underdeveloped regions. This is because, even though the cost of living and productivity are lower in the South compared to the North, these regions are subject to the same wage levels, which is a burden for businesses in the South. This process hinders the development of the more fragile regions and undermines the achievement of economic convergence of south to north (Faini 1999, p. 16).

Most of the negotiations between unions and business organizations took place in the North and so, most of the labour agreements have been signed in the northern regions. Employer organizations and unions have always been more active in the North than in the South, that had no choice but to passively accept these regulations. As a result, since a centralized system ensures that the same contract is extended to the south, the rules and the contracts, based on the economic needs of the northern regions, applied nationwide. By the consequence, most wages in national contracts are set close to the northern local labour market levels. Consequently, since the southern market level wage is lower than the northern one, these contracts result inappropriate and creates allocative inefficiencies. In fact, southern firm suffer for a high labour cost, while northern firms have no difficulties to sustain it. For example, 67% of national contracts were signed in Lombardy, Emilia Romagna, Veneto, and Piedmont (northern regions), while only 10% of these were signed in the South (Boeri, Moretti, Ichino and Posch 2021).

Despite big geographic differences in productivity, infrastructure, location, public goods and efficiency of local institutions in different regions, firms must face a similar minimum wage, due to national contract. Considering that South is less productive than

[6] Italy (but also France and Spain) exhibits high (also excessive) level of guarantees to the workers. The recent attempt of the Italian government to weaken the legislation on employment protection (thus increasing the possibility for employers to dismiss employees) would imply a support by workers for Italian unions. (Checchi and Lucifora 2014)

North (Chapter 4), the market clearing levels are higher in North than in South. Thus, since the minimum wage, for both north and south, is settled on northern market level, southern firms pay more than the actual productivity of the labour per worker. Analytically, assume that a Southern firm can absorb a certain labour cost LC. Without minimum wage level:

$$LC = w^* n^*$$

where w^* is the market clearing wage and n^* the number of market clearing employees' level. If we consider that the firm is bounded by a minimum wage, the total cost of the firm will be:

$$LC = \overline{w}\,\overline{n}$$

where $\overline{w}$ and $\overline{n}$ are minimum wage and number of employees under minimum wage. Wage level is above the market clearing level in South Italy: $\overline{w} > w^*$. Therefore, in order to have the same labour cost LC, the firm has to hire less: $\overline{n} < n^*$. That means that, if minimum wage is higher than market clearing wage, the firm has to hire less employees[7], in order to have the same level of total cost production. Having less production factors, firm production capacity will be reduced. By the consequence, production is limited, and employment rate does not reach its potential level, namely there is misallocation.

There are also some decentralized agreements, but they are limited to fix higher minimum wage, that basically does not change the level of rigidity of the system. Moreover, there exist opening clauses, but they are very difficult and rare to apply[8]. These clauses can be activated under specific conditions (such as crises or recessions), allowing for the modification of certain terms established in the collective agreements. As such, they appear to be an important tool for companies to handle with economic difficulties. However, since they are applicable only in very extreme cases, opening clauses are not very relevant and do not significantly help to make the system more flexible and responsive to potential shocks, even in situations of economic crisis. So, Italian companies do not have much flexibility in lowering wages, even in times of crisis.

[7] This is a very simple analytical framework. Being aware of the complexity of the labour market, we have just formalized the previously presented concepts analytically to make them clearer and more understandable. In the next chapter we formalise theoretical framework in a more rigorous manner.

[8] Two example of opening clauses are "Art. 5 – Intese modificative del CCNL" and "Art. 2. – Contrazione temporanea dell'orario di lavoro" from "Contratto collettivo nazionale di lavoro 5 febbraio 2021 per i lavoratori addetti all'industria metalmeccanica privata e alla installazione di impianti". https://www.fiom-cgil.it/net/images/CCNL/INDUSTRIA/2021_02_05-CCNL-federmeccanica.pdf

Therefore, even if there is not a minimum wage, Italian firms cannot pay a wage below the level set by national contract, regardless from their productivity and product demand. Private sector Italian firms have a limited degree of wage flexibility, since national contract allows a low geographic differentiation.

Since the 1990s, there have been attempts to make the system more flexible. For example, proximity agreements ('Accordi di prossimità') in the reform D.L. 138/2011 (art.8) were one of the most significant reform aimed at introducing flexibility in the system. However, these agreements did not lead to make the wage system more flexible, because this type of reforms appears to favour decentralization, but in an uncoordinated manner.

In fact, although wage coordination rules are generally respected, many company agreements provide for fixed wage increases, in contrast to rules requiring increases linked to productivity and economic performance. This process leads to a progressive ineffectiveness of flexibilization policies, because they are manoeuvres that, although decentralising, appear uncoordinated (Tomassetti 2017, p.1).

Thus, despite the attempts, Italy has never truly been able to make wages more flexible and therefore link them to productivity and wages remain homogenous. Wages are uniform between North and South, especially in the public sector. The salaries of employees working in the public sector are paid equally for each province according to the specific task and in proportion to the years of experience.

Table 1. Homogeneity of Italian nominal wages.

	North	Centre	South
Bank employee	1666	1667	1664
Energy company employee	2736	2923	2931
School teacher	1305	1305	1305

Note: The table presents the median monthly wages for three different categories of workers. The last row shows the equality of wages among civil servants. Source: Based on data from Boeri, Moretti, Ichino & Posch (2021, p.9)

Table 1 shows the median monthly salaries for private sector (bank teller and energy company) and public sector (elementary school teacher) for North, Centre and South. The first line shows the median salary for bank employees with 10 to 20 years' seniority, in an occupational level 6 position, for the years 1993-1995, with sample sizes

of 423 (North), 140 (Centre), and 105 (South). In the second row, there are the median monthly salaries for a permanent contract in a large Italian energy company in 2016, with sample sizes of 91 (North), 64 (Centre), and 130 (South). The third row presents the salary of an elementary school teacher after 5 years in 2014. One can notice a strong homogeneity in salaries.

Specifically, in bank teller sector the salaries vary by only 1€ between North and Centre (0.06%), 2€ between North and South (0.12%) and 3€ between Centre and South (0.18%). In practice, wages are the same in every part of Italy. As for energy companies, the difference is a bit higher than bank teller, but still not significant. North median salary is even lower than Central and Southern median wages. Wage in North are the 6.40% lower than the Central one and 6.65% lower than the Southern one, in this energy company. The difference between Central and Southern wage is only the 0.27%. Thus, there is a minimal difference in wage. In the public sector this homogeneity is even more accentuated, since the salary for an elementary school teacher is the same throughout Italy (1305€).

Over the decades nominal wages among workers have become slightly more heterogeneous. Nonetheless this phenomenon does not concern differences between macro regions, but rather differences between job titles and skilled/unskilled workers, which reflect the sectoral minimum wages set by collective agreements (Devicienti, Fanfani, Maida 2019, p.26).

In synthesis, national contracts cover almost the total labour, and the bargaining system is highly centralized and rigid in Italy. The primary consequence is a strong wage uniformity between the northern and southern regions, regardless of their productivity. Wage level does not reflect productivity, and the southern firms have to sustain an excessive labour cost. To maintain the same level of total costs, southern companies cannot hire as many workers as northern companies, creating spatial misallocation. The consequence of this is a high rate of non-employment and inefficiencies in local labour markets.

2.2 Germany

Germany offers an interesting case study to compare with Italy, because wage bargaining system is totally different. It has a minimum wage to 12.41€ per hour from 2024[9], but allows, anyway, flexibility to firms in all the regions ("Länder") because minimum wages align on local market level, thus making the system less rigid. The German wage setting system is based on sector level collective agreements that arise from negotiation between union and employers' representatives at regional level. The principle of double affiliation[10] applies, which requires either employers or workers to belong to the associations that signed the agreements. In this way, the contracts do not include workers and employers who are not members of the unions and organizations that signed (OECD 2017, p. 140).

German agreements are suffering an internal and external erosion. External erosion refers to the tendency for a decrease in contracts and negotiations between employers and workers, while internal erosion, which also reinforces external erosion, refers to the gradual weakening of the relevance of existing agreements. Even though 63% of workers in West Germany and 43% of those in East Germany are covered by area-wide wage agreements, there are many cases of remuneration below the levels set by collective agreements. (Ochel 2005, p.6).

This is plausible because German companies more frequently exercise opening clauses than Italian companies. These clauses, which were also extended to wages starting in 1995 in Germany, allow companies to lower wages under more flexible conditions compared to the terms of Italian clauses (Brändle et al. 2011, p.1). In fact, while before only unions could authorize a revision of the clauses, now company representative councils can modify the terms of the clauses with or without union involvement, according to Pacts for Employment and Competitiveness (PECs), which spread significantly in Germany (Seifert and Massa-Wirth, 2005 p.1).

This enables German firms to answer faster to the challenge of the global competition and the preserve employees from firing. German system is much more reactive to external stimulus, while the Italian economy is very static. The frequency of

[9] Minimum wages are differentiated by sector. 12.41 euros is the lowest, 20.86, for educational staff with additional qualifications, is the highest. Federal Ministry of Labour and Social Affairs shows the Statutory agreed branch-specific minimum wages in Germany (DeStatis, Statistische Bundesamt)

[10] The principle of double affiliation in the context of union bargaining refers to the practice where a worker is simultaneously affiliated with two different unions. This principle can impact the dynamics of collective bargaining, as a worker might be represented by more than one union in separate negotiations or could benefit from the protections offered by both unions.

opening clauses is elevated: half of German companies (53% in the West, 50% in the East) make use of them (OECD 2017, p.153).

The necessity of high flexibility and opening clause emerge in the post-unification Germany of nineties, especially in East post-DDR Germany. It was necessary to decrease the coverage of sectoral bargaining agreements and a shift to more widespread company-level agreements, in order to keep German manufacturing industry competitive at international level, especially against eastern Europe, that has a lower cost of labour.

So, German workers are less represented than Italian one in regional bargaining, especially in East. The decline in the coverage of collective bargaining at sectoral level and the increase in opening clauses are quantified in Table 2.

Table 2. German workers under collective bargaining and opening clauses.

	% Industry contract		% workers subject to opening clauses	
years	West	East	West	East
1996	69.22	56.30	-	-
1998	67.77	50.46	-	-
2001	63.11	44.60	-	-
2003	62.08	42.58	-	-
2005	58.74	41.89	33.36	23.69
2007	56.18	40.57	38.30	28.19
2009	55.46	38.35	-	-
2011	53.70	37.44	47.27	40.01
2013	52.03	35.13	-	-

Note: This table presents the percentage of German workers (both east and west) under union contracts and opening clauses. The study is based on research from 1996 to 2013. Source: Based on data from Boeri et al. (2021, p.9).

One can notice that the percentage of workers under industry contract decreases through the years in general, and eastern workers are much less represented than western ones (by 17 points on average). On the other hand, the percentage of workers subject to opening clauses are increasing, and west shows a higher percentage than East.

Adding to this discussion the fact that wage agreements are made at the regional rather than the national level, it becomes clear that Germany benefits from a flexible and decentralized wage-setting system. For this reason, German companies are better able to align wages with productivity and a large wage inequality rises. Since companies across

Germany can pay labour costs in line with the local market level, they can optimize both production and employment numbers and it does not face employment issues and regional misallocation.

2.3 European Union countries

Each European country has its own institutions that rule labour market. It is difficult to compare different systems in terms of specific competences, due to the complexity of labour market policies. However, comparing the key features of different systems helps to categorize them. The most important characteristics are the grade of centralization, the grade of flexibility and the coordination of labour market reforms. In fact, countries with rigid and centralized systems are trying to shift towards a more flexible and decentralized configuration, but reforms and economic policies need to be coordinated to achieve this goal, otherwise they risk being ineffective, as in the case of Italy.

Countries with flexible systems, like Germany, tend to leave room for company-level bargaining and allow deviations or exclusions from sectorial agreements (opening clause) in a wide range of circumstances. This type of system, called "organized decentralization", provide flexibility to the wage-setting system without having negative effects on employment (OECD 2018, p. 35). Organized decentralization can take two different forms: in the first, national sectoral agreements define general conditions but leave companies ample room to establish contract options, as in Scandinavian countries and the Netherlands. In the second, national agreements allow for deviations at lower levels from wage contracts through opening clauses, as seen in Germany and Austria (OECD 2017, p.149).

Lots of European countries are adopting the “two-tier bargaining”, in which bargaining is harmonised with the market in an industrial area. Two-tier bargaining is a system where wage negotiations at plant-level are complemented by those at the industry or national level. This system emerged as a response to pressure for greater wage decentralization from multinational and export-oriented companies. Two-tier bargaining was adopted to reconcile macroeconomic stability with greater microeconomic flexibility, linking wages to productivity, at least in the intentions. The proliferation of this type of two-tier bargaining was evident in Belgium and Denmark in the 1980s and 1990s, but, in the early 2000s, two-tier bargaining structure is in place also in the other countries,

including surprisingly also Italy, which, however, only lasted effectively until it entered the eurozone (Boeri 2014, p.6).

Although the Italian system has the characteristics of a centralised and rigid system, it has adopted the 'two-tier bargaining' structure. This shows that even if some countries have some wage bargaining arrangements in common, this does not mean that they have the same characteristics. This is to say that it is always complex to compare wage bargaining systems in their entirety. Thus, even if one finds some collective bargaining modalities in common with flexible and decentralised systems, the literature indicates that the Italian system is clearly centralised and thus rigid.

Countries with centralized and rigid systems, like Italy, France, Iceland and Slovenia, are the ones where national sectoral agreements play a dominant role, and deviations are not possible (except extreme cases) or are only allowed for wage increases compared to sectoral agreements. The collective bargaining systems of these countries play an important role and have a significant impact on wage contracts with the addition of administrative extensions to increase worker coverage. At the same time derogations, such as opening clauses, are limited and rare.

Attempts at decentralization have been made, but they are weak and uncoordinated, rendering these efforts ineffective. For example, France tried to move a more decentralized model, but the reform was stopped by a strong union opposition. Italy tried to flexibilise its system as well, but due to the lack of coordination of labour market policies, it never managed to have the same ability to link wages to productivity as the countries of organised decentralisation. The rigidity and centralization of these systems are considered the most important weaknesses of these economies, obstructing a prompt adjustment of wages in response to crises and rising unemployment. (Afonso, Dorigatti, Molina and Tassinari 2022, p. 1).

The current trend is an attempt by all European Union countries to make their systems more flexible, in order to better link them to productivity and be more dynamics to crisis. During the Eurozone crisis, wage-setting regulation faced the greatest pressure from the European Union in Mediterranean countries[11]. Greece, Portugal and Spain, for example, have moved from a centralized wage setting system to a decentralized wage

[11] In reality, Spain and Portugal started this process in the nineties, but they have never been strongly decentralized before 2015. They adopted gradually the organised decentralisation. Greece, instead, abruptly transformed its system into decentralized to cope with a severe crisis in terms of unemployment.

setting system[12]. These countries also suffered a *Great Recession* in 2002–2014, that changes labour market conditions and economic growth[13]. So, the challenge of government is to lead the labour market in that direction[14]. Countries adapt the structure of their collective bargaining systems to adjust to supply shocks (Flanagan 1999, p.2).

The European Union itself is also actively pushing countries to move towards organized decentralization. Since 2008, the EU has been pressuring member states to "marketize" their collective wage-setting mechanisms. These pressures have led to significant changes in some countries (such as Spain, Portugal, and Greece) and have had little effect on others (Italy, in particular). The changes have ranged from the decentralization of bargaining in some countries to the imposition of radical changes under the Memorandums of Understanding (MoUs) by European and international institutions in countries (Greece, Ireland, Portugal, Romania, Spain, and Cyprus) that received financial assistance packages (Marginson and Welz 2015, p.13).

Since Italy is a country with a system that show lots of similarities with the old Mediterranean ones, it is legitimate to wonder whether it is appropriate for Italy to modify its wage setting system, adopting the organised decentralisation. Italy lags to the other Mediterranean countries, because it started this process later and the policies are not coordinated properly. Despite some laws being enacted[15], the wage-setting system remains unchanged in its basic structure. The decentralization process has only just begun for Italy, but there is still a long way to go.

[12] In recent decades, also Germany has adopted decentralisation of collective bargaining in order to stem rapid wage developments that centralised would not moderate (Ochel 2005, p.1).

[13] After 2008 crisis, economic conditions of households have had a bigger impact on fertility than in the pre-crisis period, probably due to the accumulation of multiple negative economic developments and increased uncertainty. This determines a further brake on the population growth rate and thus on economic growth (Matysiak, Sobotka and Vignoli 2020, p.14).

[14] The new European labour market interventionism is based on overcoming all rigidities, including excessive trade union power, that prevent downward wage flexibility (Schulten and Müller 2012, p. 23).

[15] National collective agreement of 22 January 2009: Introduction of a general opening clause for wage regulations deviating from sectoral agreements at company level (the agreement was not signed by the largest Italian trade union federation CGIL). National collective agreement of 28 June 2011: All sectoral agreements shall contain opening clauses, according to which at the enterprise level there may be deviation from sectoral standards under certain circumstances. Law No. 148 of 14 September 2011: Company collective agreements can deviate downwards from sectoral agreements and certain labour law provisions. (Schulten and Müller 2012)

3. Theoretical Framework

This chapter develops the Boeri's theoretical model, describing the dynamics between the system and wages, employment and level of prices. The first assumption is the existence of a difference between two regions of the same country n and s in terms of productivity, because Italy and Germany have important internal difference in productivity across different regions (Chapter 4). Northern Italy and western Germany are much more productive than south and east, respectively. Therefore, we assume two regions $r = \{n, s\}$ which differ in productivity level (n is more productive than s):

$$A_n > A_s$$

This assumption reflects the similar productivity distribution of Italy and Germany. It is crucial to consider also the difference in wage setting system, comparing spatial equilibria under two different scenarios:

- Free market scenario (Germany): Local nominal wages can freely adjust to reflect local productivity.
- Fixed wage case (Italy): Local nominal wages cannot adjust due to institutional constraints, but workers and firms can relocate freely.

These two scenarios are extremizations that do not fit precisely with German and Italian systems. Clearly, no nation has such an extreme approach on wage setting mechanism. However, these extremizations simplify our theoretical analysis, and they are useful to understand the dynamics of the labour market depending on the system. So, we can rely on these two scenarios for the purposes of the research.

In the first scenario, which closely resembles the German system but remains an idealized version, there is full employment in both regions because there are no rigidities preventing the labour market from reaching equilibrium. As a result, in equilibrium, real wages are equal across the two regions, enabling the system to reach an efficient allocation. In the second scenario, idealized as well, but similar to the Italian system, wages are set by a national contract, $\overline{w}$, in both regions. Here, employees receive the same salary regardless of their region, making wages exogenous rather than endogenous.

The analysis is based on Rosen-Roback[16] spatial equilibrium model. The cases in which labour market does not reach equilibrium are not considered in this analysis. In this model, companies maximize their profit functions, all workers are renters, labour supply is fixed, assuming zero mobility costs, and there are no heterogeneous preferences for location. Additionally, another assumption is that the two regions are identical ex-ante, except for their Total Factor Productivity (TFP) levels[17].

3.1 Configuration

Consider two regions $r = \{n, s\}$. From the perspective of firms, the starting point is a Cobb-Douglas production function:

$$Y_r = A_r K_r^{(1-\alpha)} E_r^{\alpha} \tag{1}$$

where Y is the output, A TFP, K capital, E number of employed and α is a parameter showing the elasticity of output with respect to labour. Population of each region is L_r, with the total population of the country $\bar{L} = L_n + L_s$ assumed fixed. The two regions are ex-ante identical apart productivity: $A_n > A_s$.

Moving to the workers' perspective, the utility of a representative worker in region r is given by:

$$\Omega_r = \frac{w_r}{p_r^{\sigma}}(1 - u_r)^{\delta}$$

where w wage, $u_r = 1 - (E_r \backslash L_r)$ unemployment rate, p_r price level, σ indicates the sensitivity of utility to price changes and δ measures the elasticity of value with respect to the unemployment rate. In equilibrium, workers are indifferent across the two regions. We assume that all workers are renters, labour supply is fixed at L, we assume

[16]The Rosen-Roback is a model in urban and regional economics that describe how different geographic areas reach the equilibrium. Wages, housing prices, and population levels are the main factors that the model consider. Workers choose in which region live based on trade-off between the level of wages, the cost of living (primarily housing), and amenities in different locations. Firms choose locations based on labour costs (wages) and the productivity advantages of different areas. The equilibrium ensures that firms and workers are indifferent to moving to another location due to similar profit levels across regions (for firms), wage level, cost of living and amenities.

[17] Namely, their efficiency with which all inputs (capital, labour, materials, etc.) are used together in the production process. As we said before, region n is more productive than s: $A_n > A_s$.

zero mobility costs and no heterogeneous tastes for location. In equilibrium workers are indifferent across the two regions: $\Omega_n = \Omega_s$

Firms optimally decide how many workers to employ and how much capital to use. Similar to the standard model, factor demand depends on first-order conditions, meaning that the price of each factor must equal its marginal product:

$$\max_{L_r K_r} A_r K_r^{1-\alpha} E_r^{\alpha} - w_r E_r - r_r K_r$$

where w_r represents the wage, the cost of labour per unit of labour, and r_r is the capital return rate, the cost of capital per unit of capital. Solving this maximization problem (with FOC's), we obtain the demand for labour and capital (calculations in Appendix A):

$$w_r = \alpha\, A_r\, K_r^{(1-\alpha)} E_r^{(\alpha-1)} \quad \text{and} \quad r_r = (1-\alpha)\, A_r\, K_r^{-\alpha} E_r^{\alpha} \tag{2}$$

Taking the logarithm, we get:

$$\ln w_r = \ln \alpha + \ln A_r + (1-\alpha) \ln K_r - (1-\alpha) \ln E_r$$

$$i_r \equiv \ln r_r = \ln(1-\alpha) + \ln A_r - \alpha \ln K_r + \alpha \ln E_r$$

where $i_r \equiv \ln r_r$ is the interest rate in r. This equation must be equated with the capital supply. The quantity of capital is directly proportional to the price at which it is supplied to companies: $ir = \mu \ln K_r$. In addition, the more people require housing, the more expensive the housing: $\ln p_r = \gamma \ln L_r$ (housing costs are proportional to regional population). $wn, ws, En, Es, Kn, Ks, pn, ps, Ln, Ls, in, is$ are all endogenous variables in free-market case. In fixed wage case, instead, wages are exogenous.

3.2 Free-market scenario

The first scenario that implies free market with high grade of flexibility, where wages always reach the clearing market level. In equilibrium, workers are indifferent in which region to stay, because they can find a job in both regions, and the real wage would

be the same. In fact, since there are no rigidities, the labour market balances, and both regions are in full employment: $u_n^* = u_s^* = 0$

$$u_r^* = 0 \rightarrow E_r = L_r$$

Therefore, the equilibrium condition ($\Omega_n = \Omega_s$) is:

$$\frac{w_n^*}{p_n^\sigma} = \frac{w_s^*}{p_s^\sigma}$$

Applying logarithm to both sides:

$$\ln w_n - \sigma \ln p_n = \ln w_s - \sigma \ln p_s$$

In equilibrium, $i_r \equiv \ln r_r = \mu \ln K_r$. In addition, we assume that housing supply is upward sloping: $\ln p_r = \gamma \ln E_r$. Given these assumptions and conditions, one can find employment, population, capital, and housing prices of both regions. The logarithm differences between the two regions are (calculations in the Appendix A):

$$\ln E_n - \ln E_s = \frac{1+\mu}{\sigma\gamma(\mu+\alpha)+\mu(1-\alpha)} (\ln A_n - \ln A_s) > 0 \quad (3)$$

$$\ln K_n - \ln K_s = \frac{\sigma\gamma+1}{\sigma\gamma(\mu+\alpha)+\mu(1-\alpha)} (\ln A_n - \ln A_s) > 0 \quad (4)$$

$$\ln w_n - \ln w_s = \frac{\sigma\gamma(1+\mu)}{\sigma\gamma(\mu+\alpha)+\mu(1-\alpha)} (\ln A_n - \ln A_s) > 0 \quad (5)$$

$$\ln p_n - \ln p_s = \frac{\gamma(1+\mu)}{\sigma\gamma(\mu+\alpha)+\mu(1-\alpha)} (\ln A_n - \ln A_s) > 0 \quad (6)$$

Initially, in region n, where the TFP is higher, employment, capital, and nominal wages, in equilibrium are higher than in s. This can be clearly observed by expressing equilibrium employment, capital, and nominal wages as functions of the model's external parameters that are positive ($0 < \alpha < 1$). Due to the higher TFP in n, profit-driven firms opt to hire more workers and utilize more capital in that region. The disparities in labour and capital inputs directly correspond to the differences in TFP. Housing expenses are

also higher in n than in s, as more workers reside there at equilibrium (so n has bigger population). The divergence in housing costs between the North and South regions must be substantial enough to make workers indifferent between the two areas, according to the assumption of spatial equilibrium. Consequently, while nominal wages are higher in n at equilibrium, real wages remain equal in both regions. Summarizing, compared to region s, region n is more productive, has a larger population, possesses more capital, boasts higher nominal wages, but experiences a higher level of housing prices. Therefore, real wages are equal in both regions, and there is full employment in both, even if a region n has a bigger population.

3.3 Fixed-wage case

Then, it comes the analysis of the situation in which wages are set by National Contract. One assumes that wages are set by National Contract $\overline{w}$ in both regions. Now employees have the same salary no matter in which region they live. It means that wages are exogenous and not endogenous anymore. That is a significant difference with the previous case, because wages are set on the level of region n:

$$\overline{w} = w_n^* > w_s^* \tag{7}$$

Once wage is established by the national contract, employment, non-employment, population, and housing prices adjust in the two regions. In n there is the same situation of the free market case ($E_n = L_n \rightarrow u_n = 0$), because the wage set by national contract is the same than the one in free-market case (market clearing level): $\overline{w} = w_n^*$.

However, there is a different situation in region s, since current wage is higher than the free-market case $\overline{w} > w_s^*$. This has an important consequence for firm in region s, because they must sustain a higher cost for the salary than the one in the market. By the consequence, they can assume less employees simply because labour demand is downward sloping.

Therefore, the main difference compared to the free-market scenario is that a national contract leads to lower equilibrium employment, capital, and output in region s and this has important that we discuss later.

Firms maximise their profit function by choosing the optimal quantity of capital and labour to use in the productive process. So, the procedure looks the same as the free-

market case, at least in the first calculation. The difference emerges when one considers the equilibrium $\Omega_n = \Omega_s$:

$$\frac{\overline{w}(1-u_n^{**})^{\delta}}{p_n^{**\sigma}} = \frac{\overline{w}(1-u_s^{**})^{\delta}}{p_s^{**\sigma}}$$

Notice that, since $\overline{w} = w_n^*$, $E_n^{**} = L_n^{**} \rightarrow u_n^{**} = 0$ and so (the double asterisk collective bargaining case):

$$\frac{\overline{w}}{p_n^{**\sigma}} = \frac{\overline{w}(1-u_s^{**})^{\delta}}{p_s^{**\sigma}}$$

From this equilibrium we get (Calculation in the Appendix A):

$$\ln E_n^{**} - \ln E_s^{**} = \frac{1+\mu}{\mu(1-\alpha)}\,(\ln A_n - \ln A_s) > 0 \tag{8}$$

$$\ln K_n - \ln K_s = \frac{1}{(1-\alpha)\mu}(\ln A_n - \ln A_s) > 0 \tag{8a}$$

$$u_s^{**} = \frac{(1+\mu)\sigma\gamma}{\mu(1-\alpha)(\sigma\gamma+\delta)}\,(\ln A_n - \ln A_s) > 0 \tag{9}$$

$$\frac{\overline{w}}{p_n^{**\sigma}} - \frac{\overline{w}}{p_s^{**\sigma}} = -\frac{(1+\mu)\sigma\delta\gamma}{\mu(1-\alpha)(\sigma\gamma+\delta)}\,(\ln A_n - \ln A_s) < 0 \tag{10}$$

$$\ln p_n^{**} - \ln p_s^{**} = \frac{(1+\mu)\delta\gamma}{\mu(1-\alpha)(\sigma\gamma+\delta)}\,(\ln A_n - \ln A_s) > 0 \tag{10a}$$

Firstly, employment is lower in s, given that $\ln E_n^{**} - \ln E_s^{**}$ is positive. It is interesting to compare this with equation (3). We notice a difference in the coefficient preceding $(\ln A_n - \ln A_s)$:

$$\frac{1+\mu}{\mu(1-\alpha)} > \frac{1+\mu}{\sigma\gamma(\mu+\alpha)+\mu(1-\alpha)}$$

The occupational gap of the National contract is greater compared to the free-market case. This means that in the case of the national contract, the difference between the two regions in terms of employment is more evident and accentuated compared to the free-market case, given the same productivity gap (see Equation 8).

Unlike the case of free market, there is no full employment in equilibrium in region *s*: $u_s^{**} > 0$. In fixed-wage case, $\overline{w}$ is greater than the level of wages in free market equilibrium in s, with non-full employment arising from the gap between wages and local productivity (see Equation 9).

In equilibrium, people from region *s* do not go to region *n* to find a job in this case, because the purchasing power for a worker in *s* is higher than that of a worker in *n*. So, it's advantageous for an unemployed person from region *s* to wait for a job in *s* rather than migrate to *n* (see equation 10). In this scenario, indeed, nominal wages are the same, but housing is more expensive in *n* (see equation 10a). Consequently, region *s* enjoys higher real wages.

Therefore, since real wages in s are higher, the unemployed remain in *s* in the hope of finding employment. In fact, the satisfaction of people in southern Italy depends not so much on the level of real wages but rather on employment status (Chapter 6). In contrast, *n* enjoys full employment because $\overline{w}$ is equal to the market equilibrium wage. Like in the case of the free market, housing costs are higher in *n* because employment and population are higher there, but unlike the free-market case, real wages are lower in *n*.

The key difference from the free-market case is that national contract implies fixed wages and lower equilibrium employment, production, and capital in *s* (see equation 8a). While the wage in *s* is higher than the free-market equilibrium, fewer workers are employed, because firms could hire less employees and total national employment and production decreases. That is inefficient and creates regional misallocation. As a result, production and aggregate labour income are lower. Imposing a fixed minimum wage that exceeds the local market clearing wage, creates allocative inefficiency that generates costs.

Finally, in equilibrium, businesses invest less in *s* than in *n*, but the capital intensity, i.e. the capital-labour ratio, is higher in *s* than in *n*. Intuitively, the loss of employment in *s* compared to *n* is greater than the loss of investment because the labour market cannot balance, but the capital market can. The model indicates inefficiencies in the system in this scenario. This case, as mentioned earlier, is attributable to the Italian system, which suffers from the same employment problems described by the model. These implications are supported by the empirical evidence (Chapter 4).

3.4 Aggregate Effects

In this subchapter, the comparison of two different scenarios is provided. In the free-market case, full employment is achieved because there are no obstacles preventing the normal adjustment of the market. When wages are set according to the National contract scenario, equilibrium is certainly reached as before, but there is no full employment since wages cannot drop to market levels in region *s*. This is because firms in region *s* pay wages above the market level. As a result, these companies will be able to hire fewer workers and therefore produce less. That has an impact on the intensity of capital as well.

Since $E_s^{**} < E_s^*$, it follows that $K_s^{**} < K_s^*$. By the consequence, differences on production emerge. In fact, at the aggregate level, production in the free-market case is higher than in the collective bargaining case:

$$Y_n^* + Y_s^* > Y_n^{**} + Y_s^{**}$$

Moreover, since capital and labour are not perfect substitutes, the total amount of capital in s is also lower in national contracts. The equilibrium capital stock in s is:

$$\ln K_s = \frac{1}{\mu + \alpha}[\ln(1-\alpha) + \ln A_s + \alpha \ln E_s] \tag{11}$$

In the fixed wage equilibrium employment and capital are higher in n than in s, compared to the free market equilibrium. But aggregate losses remain. Consequently, if the demand for labour is elastic, in equilibrium, labour income in the free-market case is higher than in the bargaining case.

$$\frac{(w_n^* E_n^*) + (w_s^* E_s^*)}{\bar{L}} > \frac{(\bar{w} E_n^{**}) + (\bar{w} E_s^{**})}{\bar{L}} \tag{12}$$

To understand why, note that we can rewrite this inequality as (Appendix A):

$$(\bar{L} u_s^{**})\bar{w} > E_s^*(\bar{w} - w_s^*) \tag{13}$$

The left side of the equation corresponds to labour income in the free-market case, while the right side corresponds to the labour income in the bargaining case.

Figure 1 represents graphically Equation 13. It shows the marginal product of labour (and thus the demand for labour) in region s, which is the axis on which we set the wage level. Points 1 and 2 represent the free-market equilibrium and the fixed-wage equilibrium, respectively. The function derives from the Equation 2:

$$w_s = \alpha A_s K_s^{(1-\alpha)} E_s^{(\alpha-1)}$$

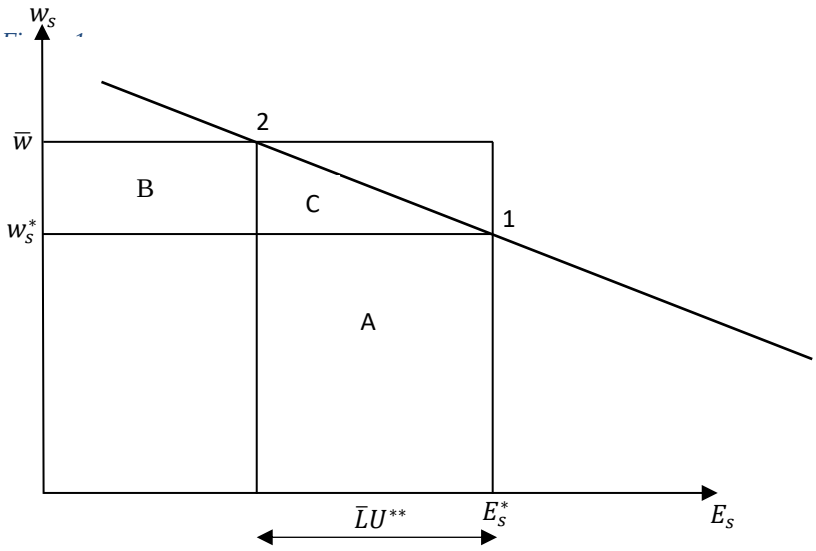
In fact, the line is the log-transformation[18] of the exponential relationship between w_s e E_s. The left side of Equation 13 corresponds to the area of rectangles A + C, while the right side corresponds to the area of rectangles B + C. Labour income is higher in the free market if area A is greater than area B.

Labour income decreases compared to the free-market case if the demand for labour is sufficiently elastic, as illustrated in the figure. For a small open economy, the demand for product and thus the demand for labour is likely elastic. In summary, national contracts that set wages above the local labour market level create rigidity in the system that leads to a loss of employment, output and labour income.

Figure 1 shows that labour income under a free market is higher than in the case of fixed minimum wages. Therefore, according to this model, setting fixed minimum wages above the local market level leads to allocative inefficiency. This is a crucial implication that emerges from the model, as it considers the wage setting mechanism a key actor that can determine labour market outcomes in geographically heterogeneous economies.

[18] We can use this formula $\ln(1-x) \approx x$ for $x \to 0$. This formula is a Taylor series centred around 0 (also known as a linear approximation). When the value of x is small, the higher-order terms in the Taylor series can be neglected, so the function behaves almost linearly, and the approximation becomes very accurate.

Figure 1. Labour demand.



Note: Labour income in the free market scenario (A+C) and in fixed wage case (B+C). Source: Based on calculations from Boeri, Moretti, Ichino and Posch (2021, p.18)

By developing the analysis on two extreme scenarios, it is clear the impact of the wage setting system that are also decisive in the economies of Italy and Germany, particularly on outputs such as the level and geographical distribution of nominal wages, employment and output. Moreover, in the Italian economy, where significant regional disparities exist, inefficient allocation results in severe economic losses in the less developed southern regions.

Next chapter shows the empirical evidence that supports these model implications, analysing data on wage distribution, employment, and regional productivity differences. Then it proceeds with the counterfactual analysis to quantify the aggregate costs of misallocation. The results obtained from the counterfactual calculations will be our quantitative measure of the Italian economic loss due to regional misallocation.

4. Empirical Evidence

This chapter shows that the assumption and the theoretical implications of the model are in line with empirical data basis and northern Italy and western Germany are more productive than south and east. It presents data on the internal differences in nominal wages, non-employment rate, cost of living and real wages. It shows empirical proof that in Germany the nominal wage is positively linked to productivity (Equation 5, Chapter 3). Then, the occupational gap of the National contract is greater compared to the free-market case (Equations 3 and 8). In addition, it exhibits that in Italy the less productive region enjoys higher real wages (see equation 10).

4.1 Data description

The analysis is made on three datasets: 'bargaining.dta', 'amenities.dta' and 'province_data.dta'. The main sources for these datasets are the Institute for Employment Research (IAB) Establishment Panel, Wave(s) 1996 – 2013, and the National Italian Statistical Office (ISTAT), the newspaper "Sole24ore," and the "Osservatorio Mobiliare Italiano" for Italy[19].

The key results of data analysis are presented in Table 3. Starting with Italy, value added[20], average nominal hourly wage, employment rate (corrected for informal work) and the cost of living index are higher in the North with respect to South. As for Germany, productivity is higher in the West, as is the daily nominal wage and the cost of living, while employment rate is slightly higher in East. Internal difference in productivity is quantitatively similar between the two countries (11471€/worker for Germany; 9394€/worker for Italy) and the local distribution of the CPI is the same — 11 points for both countries.

Focusing instead on nominal wages and employment rates, there are significant differences between Italy and Germany. In fact, in Italy, wages are relatively homogeneous between the North and South (only a 0.3€ difference), while in Germany there is significant wage inequality between East and West (a difference of 23.6€). The opposite is true when considering the employment rate: in Germany, there is homogeneity (with only a -0.3% difference), while in Italy, there is a significant disparity, as the

[19] Appendix B contains a more comprehensive description of sources and data performed in the analysis.
[20] Value added is a measure of productivity (see productivity (value added) 4.2).

employment rate difference between the North and South is 13.7%. Germany suffers from a large wage disparity, because wages are linked to productivity (see Nominal wage, 4.3), while Italy has a large employment disparity which leads it to a regional misallocation. Model implications align with real data.

Table 3. Synthesis of empiric data for Italy and Germany

	Italy	**South**	**North**	**N-S**
Value Added per worker	54837	49182	58576	9394
Nominal wage (hourly in €)	8.53	8.36	8.66	0.3
Employment rate corr. %	65.0	57.3	71.0	13.7
Local CPI	98.9	92.3	103.3	11

	Germany	**East**	**West**	**W-E**
Value Added per worker	52901	44059	55530	11471
Nominal wage (daily in €)	91.6	73.4	97.0	23.6
Employment rate %	72.1	72.3	72.0	-0.3
Local CPI	100	91.5	102.5	11

Note: Value-added is calculated for all sectors in each geographic area, as computed by the OECD (for Germany) and ISTAT (for Italy), and is a weighted average with respect to the number of provinces (for Italy and Germany) and an arithmetic mean (for Northern and Southern Italy, and Eastern and Western Germany). It is important to note that for Italy, we have "hourly" wages net of taxes, while for Germany, we have "daily" wages gross of taxes. The employment rate refers to the number of people aged 15 to 64 employed out of the total population in that age group. The Local Consumer Price Index (CPI) is constructed using housing prices based on the method described in Appendix C. In the first column there are the weighted mean with respect to number of provinces (41 and 62 in South and North Italy, and 22 and 74 in East and West Germany), on the second and the third the arithmetic mean. Calculations are explained in Appendix C. Data, from 2010, refers to Boeri et al. (2021)

4.2 Productivity (Value Added)

The main assumption of the model is that region *n* is more productive than region *s*. The empirical data indicates that northern Italy is more productive than southern Italy, and western Germany is more productive than eastern Germany. It should be specified that the only measure of business productivity available at a detailed geographical level in Italy and Germany is gross value added. Therefore, in this context, gross value added is used as a synonym for productivity (Appendix B). Figure 2 shows this similarity in difference in productivity per worker.

In Boeri, Moretti, Posch and Ichino (2021, p.19), the productivity internal differences percentage of both countries are $\Delta^{ita} = 17.6\%$ and $\Delta^{ger} = 22.8$. These calculations are derived from the following formulas:

$$\Delta^{ita} = \frac{A_n - A_s}{A_n} \qquad \Delta^{ger} = \frac{A_w - A_e}{A_w}$$

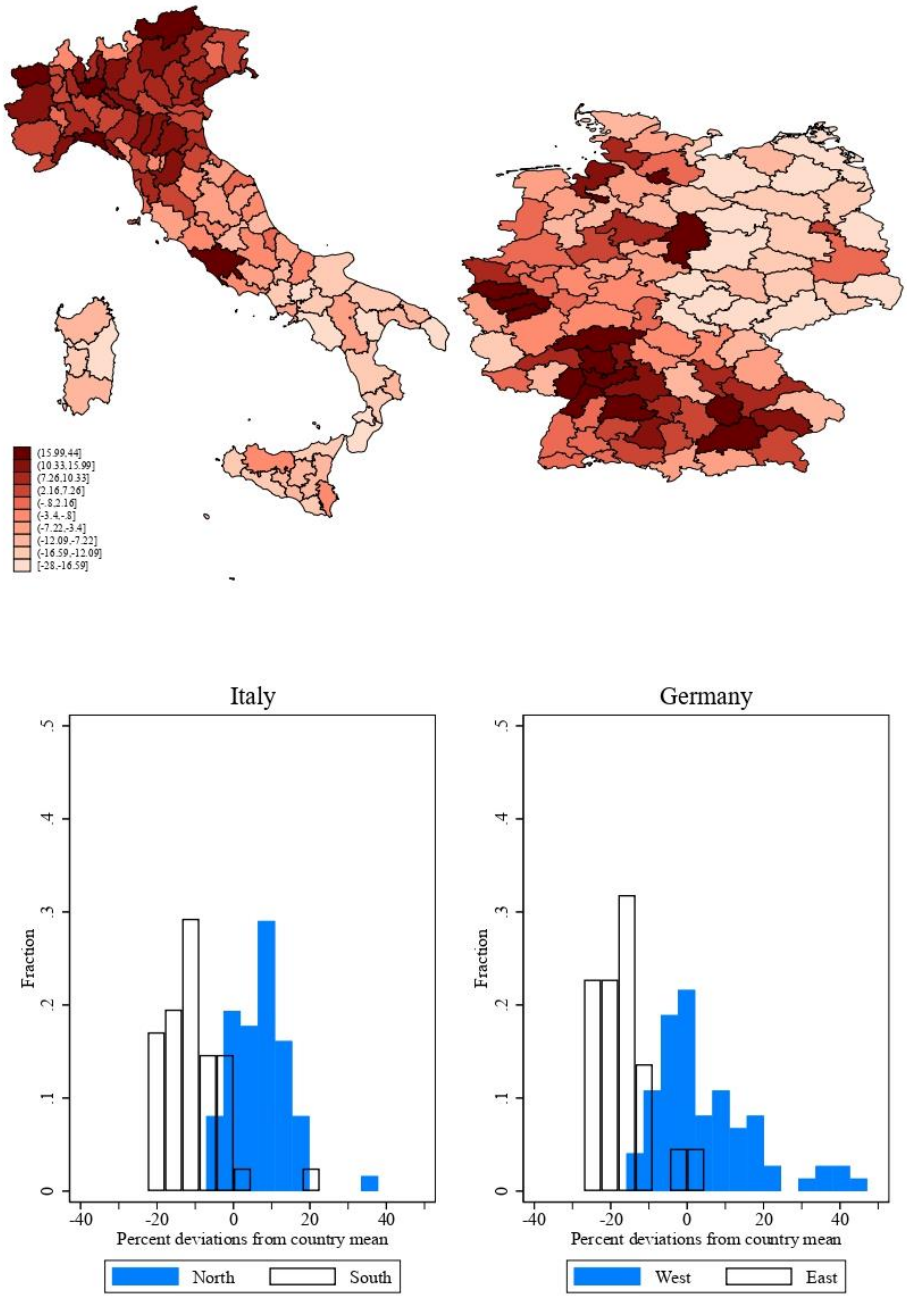
Where A_n, A_s, A_w and A_e are the added values of the median provinces of north south Italy, and east and west Germany respectively[21].

Another interesting aspect to highlight concerns the comparison between geographical distribution of German and Italian productivity among the provinces in 2010. In fact, the most of the distribution between -20% and +20% in both countries. The difference between the 90-10th percentile and the 75-25th percentile is 32.9% and 19.7% in Italy and 50.2% and 19.3% in Germany.

This variation is not atypical in industrialized countries (Hornbeck, Richard and Moretti 2018), although this geographic distribution similarity between Italy and Germany is particular because they show a split between two distinct macroregions. Italy and Germany have significant differences in their industrial and institutional structures, but the similarity in the production geographic distribution helps us understand even more precisely what the effects of a new flexible mechanism for the Italian economy might be. At this point, one can state that the assumption made in the model $A_n > A_s$ is in line with the empirical data. Moreover, these data on gross value added will be crucial in calculating the aggregate costs for Italy in the analysis of counterfactuals (Chapter 5).

[21] For what concern Italy, the value added of median provinces in the north (Pisa) and the south (Rieti) are: $A_n = 59052$ and $A_s = 48617$.

Figure 2. Mean value added per worker and value added geographical distribution.



Note: This figures plots deviations from not weighted means value added for each province in 2010. Source: Based on data from Boeri et al. (2021, p.20)

The causes of these differences are not the subject of this analysis. Economic history of the Italian Republic had a certain impact on the current economic polarity. However, it would be a serious mistake to interpret the persistence of regional disparities as if laws, reforms, policies and theoretical framework had never changed through the last decades (Iuzzolino, Pellegrini and Viesti 2013, p. 26).

In Germany, the dissimilarity of Eastern and Western regions is conceivably caused by the state-owned economy of DDR and remains significant also after decades of reunification efforts. Western Germany is characterized by higher GDP per capita, greater industrialization, and lower unemployment rates compared to the east. The institutional framework and main actors were significantly different between the two Germanies, which might lead us to think that this difference became deeply rooted and persists today. However, the values of eastern Germany are actually oriented towards the free market. In fact, the values most compatible with a market economy are strongly present among the inhabitants of eastern Germany (Van Hoorn and Maseland 2010).

It is well known that the causes of these divisions are not solely institutional and economic but also cultural, political, and historical, but since it is not the aim of our analysis, this aspect will not be further discussed.

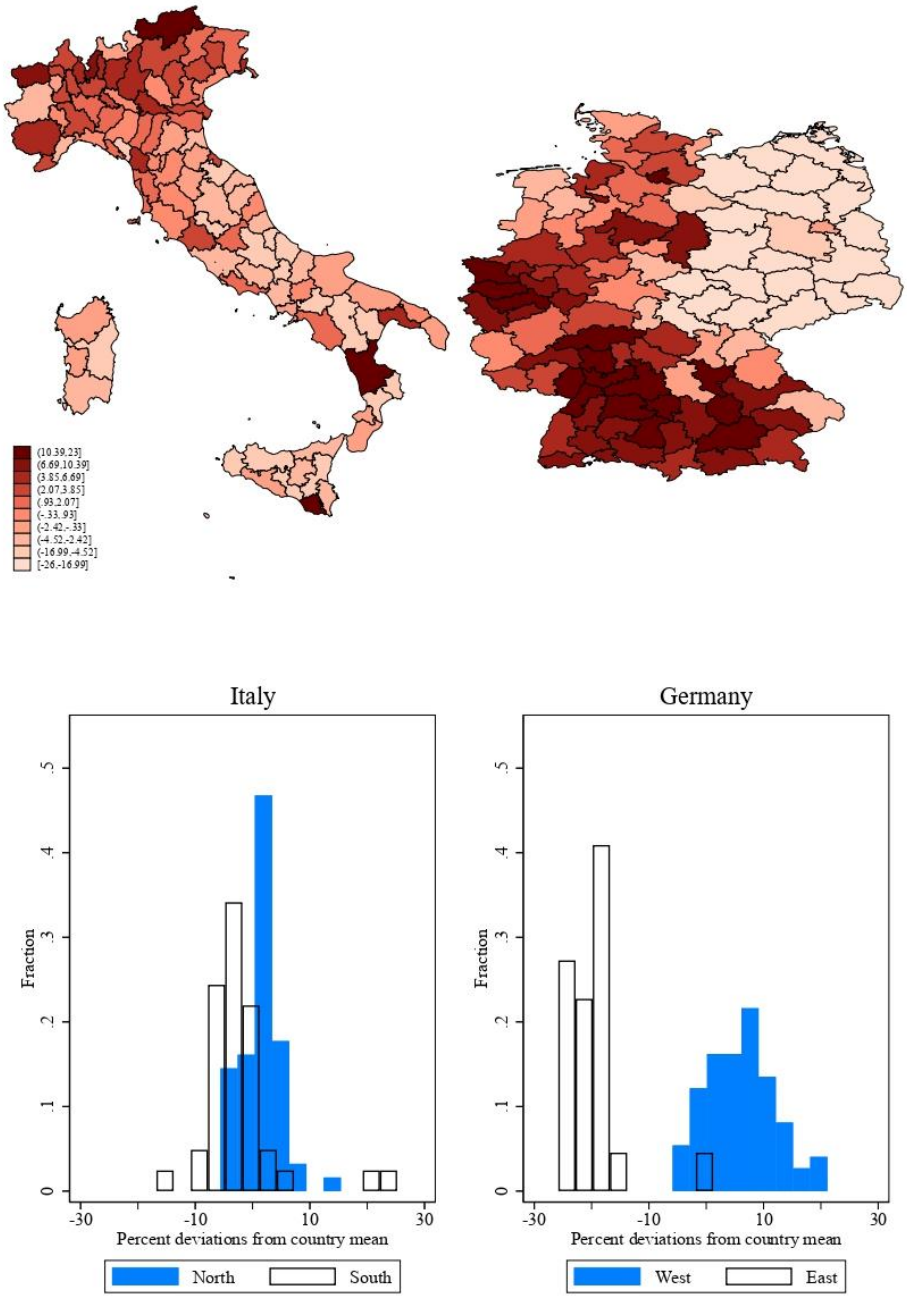
4.3 Nominal wages

Empirical data tells that nominal wages and productivity go hand in hand in Germany, while in Italy, low productivity does not correspond to low nominal wages. This is why there is a more significant wage disparity between regions in Germany compared to Italy. Figure 3 shows us the level of nominal wages by province for Italy and Germany. It can be immediately noticed that there is a very marked difference between German macro regions, while in Italy this difference is significantly less pronounced. Indeed, there are provinces in Southern Italy (Calabria, Puglia, and Sicily) that have relatively high level of nominal wages. The maps below provide a qualitative overview of the territorial distribution of nominal wages. Figure 3 shows also the deviations from country mean of the nominal wages of each province in 2010. Distribution is more concentrated in Italy than in Germany. Most of the distribution in Italy is between -10% and +10% compared to the national average. Italian mass distribution is compressed between -10% and +10% relative to the national average, while in Germany it is between -26% and +22%. The difference between the 75th and 25th percentile and the difference between the 90th and 10th percentile were 5.8% and 10.3% in Italy, respectively, and 13.0% and 43.3% in Germany.

Thus, the distributions of northern and southern Italy overlap for a large part, whereas those of eastern and western Germany are strongly separated. This tells us that nominal wages are significantly more homogeneous in Italy than in Germany. After these

considerations, the difference between Italy and Germany in terms of wage homogeneity seems evident.

Figure 3. Nominal wages distribution.



Notes: This figures plots deviations from not weighted country mean of the nominal wages in 2010. Source: Based on data from Boeri et al. (2021, p. 21)

Empirical data that will illustrate the correlation between nominal wages and value added (productivity) in both Italy and Germany. Figure 4 shows the scatter plot of

the relationship between the logarithm of the conditional mean nominal wage per province on the y-axis and the logarithm of the mean value added on the x-axis in 2010, and it shows a clear difference between Italy and Germany. In Germany, nominal wages and value added are significantly correlated, while in Italy this correlation is much less evident. These data clearly show Italy's problem in relating nominal wages to productivity, due to a very rigid wage-setting system.

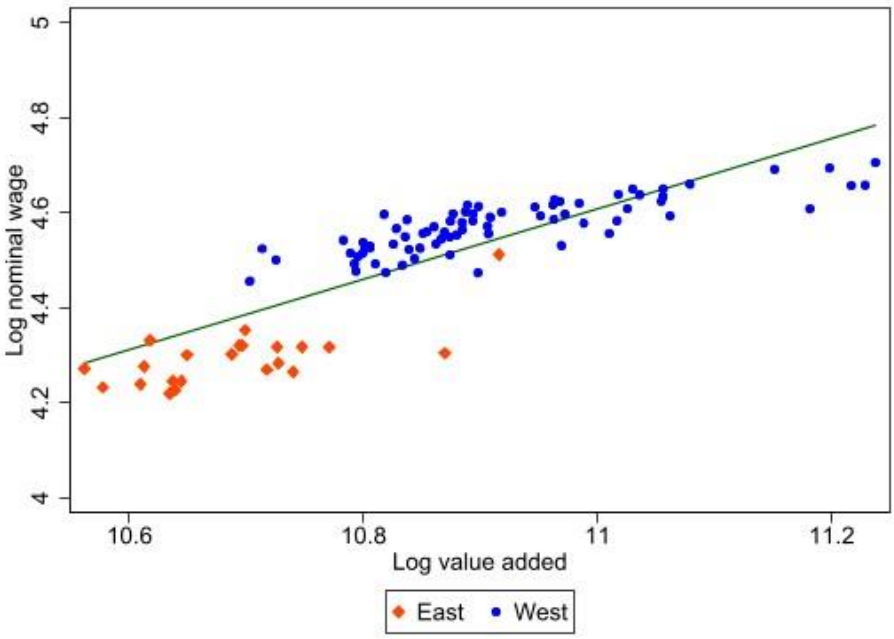
Columns 1 and 3 of Table 4 show us that the coefficient of nominal wages on local value added is approximately four times higher in Germany than in Italy (0.20 in Italy and 0.74 in Germany, respectively). In columns 2 and 4, after conditioning on regional fixed effects, the elasticity for Italy falls to 0.14. However, in Germany it is at 0.38, still three times higher. This table gives us a quantitative measure of the difference between Italy and Germany in the relationship between nominal wages and value added. In Germany, nominal wages and productivity are significantly correlated, whereas in Italy they are not.

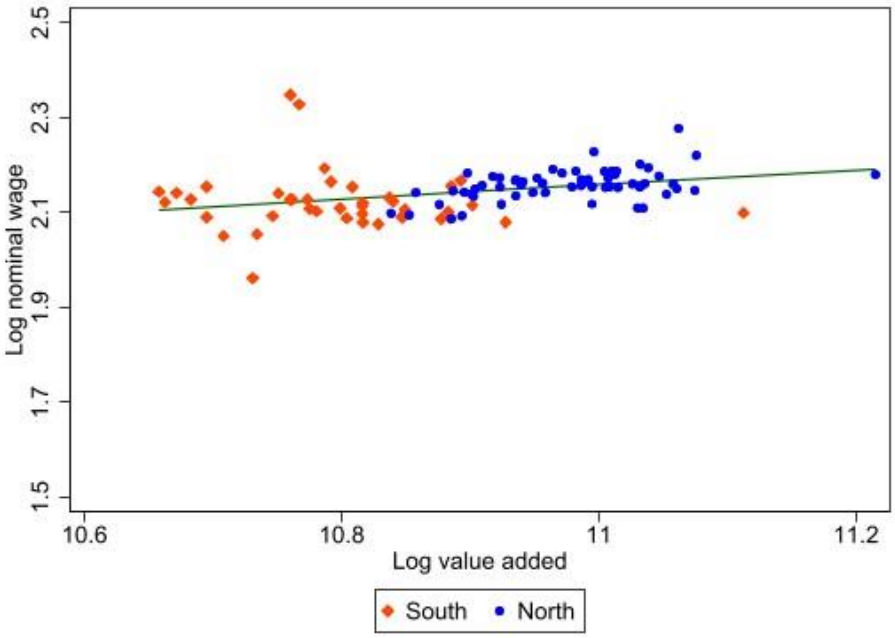
Table 4. Elasticity of nominal wage on value added for Italy and Germany.

	Italy		Germany	
	(1)	(2)	(3)	(4)
Log value added	0.195	0.137	0.739	0.380
	(0.013)	(0.019)	(0.013)	(0.008)
Region FE:	No	Yes	No	Yes
Provinces:	103	103	96	96

Note: The table presents the coefficients of the average value added in a regression of the average nominal wage on the value added for each province, in logarithm. These coefficients are the elasticity of the nominal wage to value added for Italy and Germany, with the corresponding standard errors. Columns (2) and (4) include fixed effects for Northern Italy and West Germany.

Figure 4. Nominal wage and value added.





Notes: It is shown the correlation between mean conditional nominal wages and value added in logarithm of every provinces in 2010. Dots represent provinces. Germany is represented by the figure above, Italy below. Source: Based on data from Boeri et al. (2021, p.23)

4.4 Employment rate

As mentioned, Italy and Germany are similar in terms of regional differences in productivity. On one hand, western Germany and northern Italy are much more productive than the east and the south respectively. On the other hand, nominal wages are very different between western and eastern Germany, but they are quite homogenous in Italy. This difference in the geography of nominal wages is due to the opposite features in the wage-setting systems between Germany and Italy. The wage-setting mechanism is very important because it can affect the unemployment rate as well. The key point is that in Germany, companies can pay their workers at the market price (market-clearing level). This allows German companies to choose the optimal level of labour to hire. On the other hand, in Italy, the situation is more complicated. The rigidity of the system prevents companies in southern Italy from paying the workforce at the market level. This creates a significant inefficiency in hiring workers, which then results in a high non-employment rate. In other words, since southern companies must pay wages above the market level, they are forced to bear too high costs to maintain significant production levels. Therefore, they produce less and hire less. In the north, this problem is not that evident because they can fix wage at the labour price. Northern companies can absorb the cost of wages and optimize the level of their workforce. For this reason, the north does not suffer low employment rate as much as southern Italy.

Empirical data show a clear division between north and south in terms of non-employment in Italy. The figure clearly shows that the south has much higher non-employment rates than the north. On the other hand, in Germany, non-employment rates are more similar between West and East. The German highest non-employment rates are found in certain western provinces. In general, Italy has a higher non-employment rate, 42.5%, with respect to Germany, 27.9%. It is important to note that the spatial distribution is different in the two countries (see Figure 5). In Italy, the distributions of north and south are completely separated, while the distribution of East and West Germany is fully overlapping. This indicates that the Italian unemployment rate is polarized between a high rate in the south and a low rate in the north, while the German unemployment rate is territorially homogeneous.

This means that in Germany, the non-employment rate is not correlated with productivity, whereas in Italy, there is a significant negative correlation between value added and unemployment. The scatterplots in Figure 6 provide a very clear picture of this.

The Table 5 presents the elasticity between value added and the non-employment rate. Since Italian areas with a low average value added have much higher non-

employment rates than areas with a high average value added, while in Germany the difference in non-employment rates is much smaller, the elasticity is significantly more negative for Italy than for Germany. Columns (1) and (5) of Table 5 show the corresponding values for the regression coefficients. The elasticity in Italy is -1.43, while in Germany it stops at -0.25. Columns (2) and (6) include regional fixed effects, under which elasticities decrease significantly, although the difference between Italy and Germany remains almost the same.

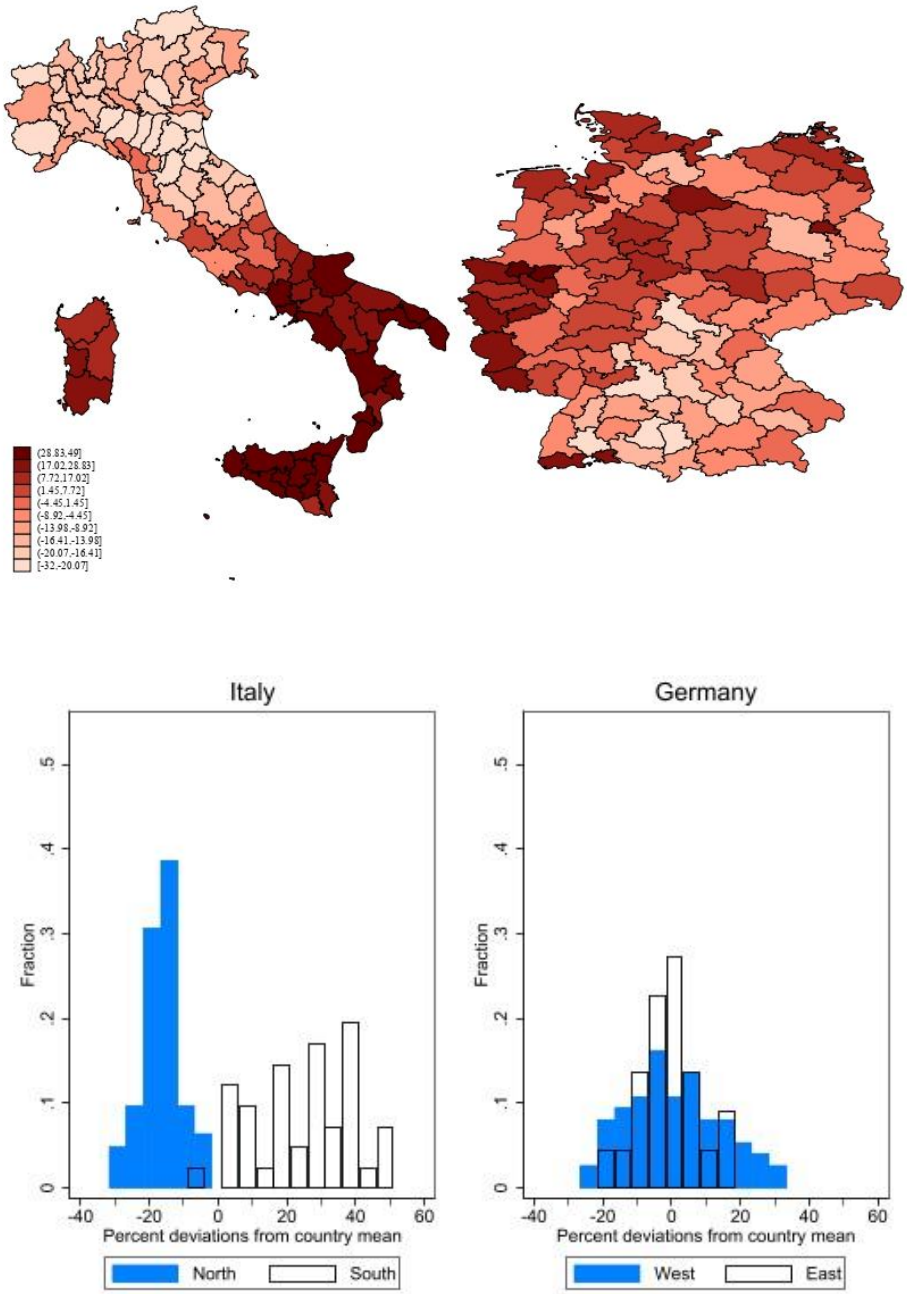
Columns (3) and (4) show the values of elasticity corrected for the informal sector. It should be specified that in Italy (especially in the South), there is a significant rate of undeclared work that must be considered. The uncorrected non-employment rate is on average 42.5% in Italy. The corrected non-employment rate is on average 34%, with a large difference of 8.5% from the uncorrected rate. That said, the elasticity of Italy between value added and the non-employment rate remains 5 times larger than the German one. In Italy, provinces with low productivity (those in the South) also suffer from a high non-employment rate. In Germany, the non-employment rate is not much correlated with value added like Italy.

Table 5. Elasticity of non-employment rate with respect to mean value added.

	Italy					
	Uncorrected		Corrected		Germany	
	(1)	(2)	(3)	(4)	(5)	(6)
Log value added	-1.434	0.534	-1.259	-0.43	-0.252	-0.110
	(0.030)	(0.031)	(0.042)	(0.050)	(0.024)	(0.031)
Region FE	No	Yes	No	Yes	No	Yes
Provinces	103	103	103	103	96	96

Note: The table presents the coefficients of the average value added in a regression of the non-employment rate between the ages of 15 and 64 on the average value added of each province, in logarithm, with the respective standard errors in brackets. the coefficients refer to data from the years 2001-2005 for Germany and 2004-2015 for Italy. the first two columns do not consider informal employment, while columns (3) and (4) take into account undeclared work, according to ISTAT data (2014). Columns (2), (4) and (6) include fixed effects for the North in Italy and for the West in Germany. Source: Based on calculations form Boeri et al. (2021, p.27).

Figure 5. Non-employment rate distribution.



Note: this figure plots deviations from the country mean of the non-employment rate of 15-64 year olds for each province in 2010. Means are not weighted by population. Source: Based on calculations from Boeri et al. (2021, p.25)

Figure 6. Non-employment rate and value added.

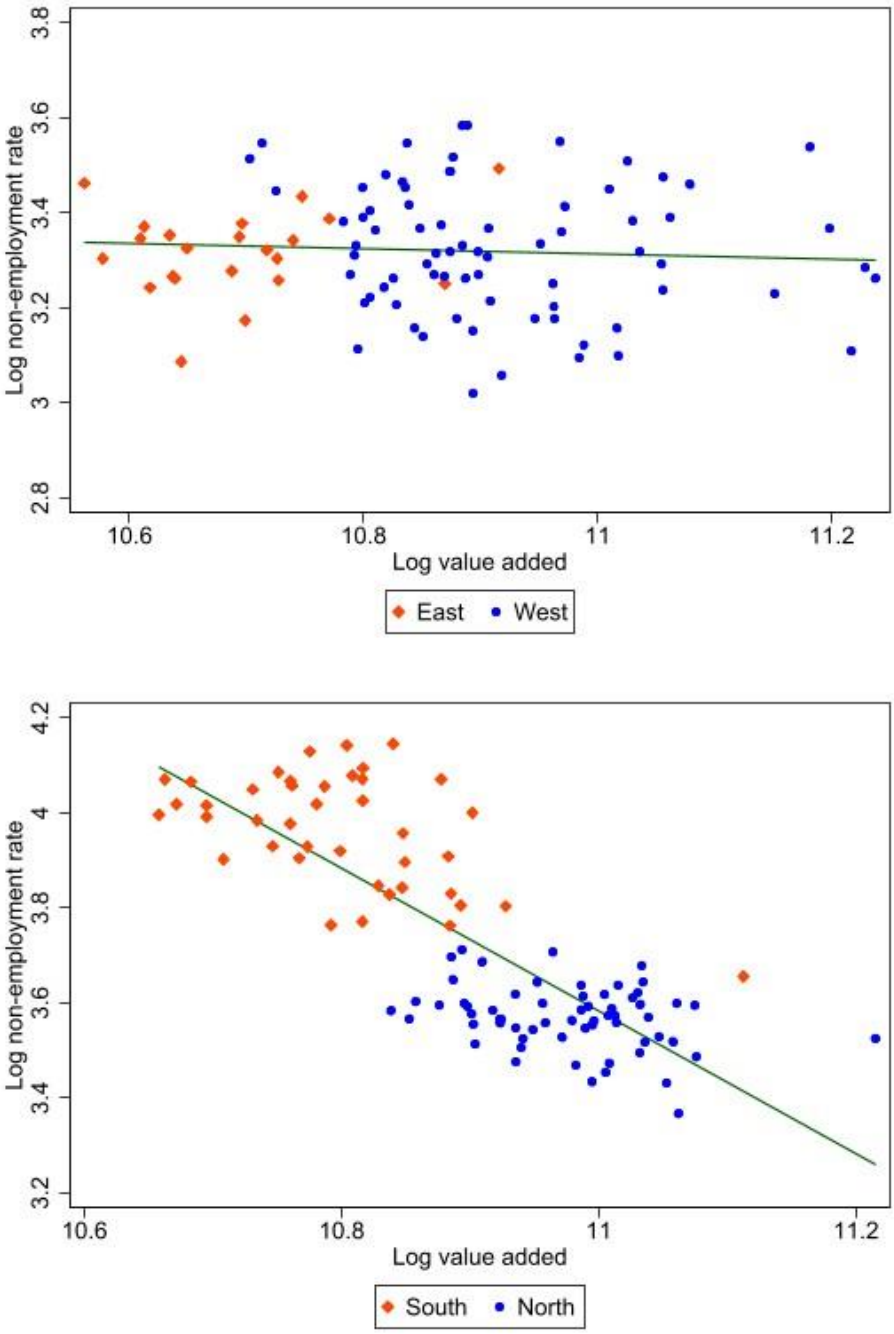


Figure 6. Non-employment rate and value added. This figure shows the correlation between log non-employment rate among 15 64 year olds and log value added across provinces in 2010. Each province in represented by a dot. Germany is represented by the figure above, Italy below. Source: Based on data from, Boeri et al. (2021, p.26).

4.5 Price level and Real wages

An implication of the model we developed in the Chapter 3 tells that the less productive region enjoys higher real wages in fixed-wage case (i.e. rigid and centralized system), due to the low price of goods in the less productive region (Equation 10). To analyse real wages, one considers both nominal wages and the cost of living. While nominal wages have been extensively studied in Nominal wage 4.3, the cost of living has not been examined yet. This subchapter presents data on the cost of living and then provide a more comprehensive overview of real wages. First, it is necessary to clarify that housing prices are considered the most important factor in the cost of living. As expected, housing prices are higher in the Western Germany and Northern Italy, as can be seen from the Figure 7. We observe from the distribution that in Germany the difference between East and West in housing prices is stronger than the difference between North and South in Italy.

The distribution of Southern Italy largely overlaps with that of Northern Italy, while the distributions of Eastern and Western Germany are slightly less overlapping. Since West and North are more populated than East and South, model's assumption $\ln p_r = \gamma \ln E_r$ aligns with empirical data. Housing cost is higher in the most populated provinces. Extending the analysis with the level of price of goods and services, we introduce the consumer price index CPI. Since the cost of housing is the most important and impactful variable of the CPI (about one third on total), geographical differences in housing costs significantly influence the local CPI, being the main factor (Moretti 2013, p.2). The consumer price index CPI measures housing prices plus the goods and services price level[22]. Geographical distribution of the CPI is similar to that of housing costs. The Figure 8 shows that, like housing costs, the CPI is higher in the north and west compared to the south and east, and that in Germany, the difference between west and east is more pronounced than Italy[23]. The consumer price index CPI measures housing prices plus the goods and services price level[24]. Geographical distribution of the CPI is similar to that of housing costs. The Figure 8 shows that, like housing costs, the CPI is higher in the north

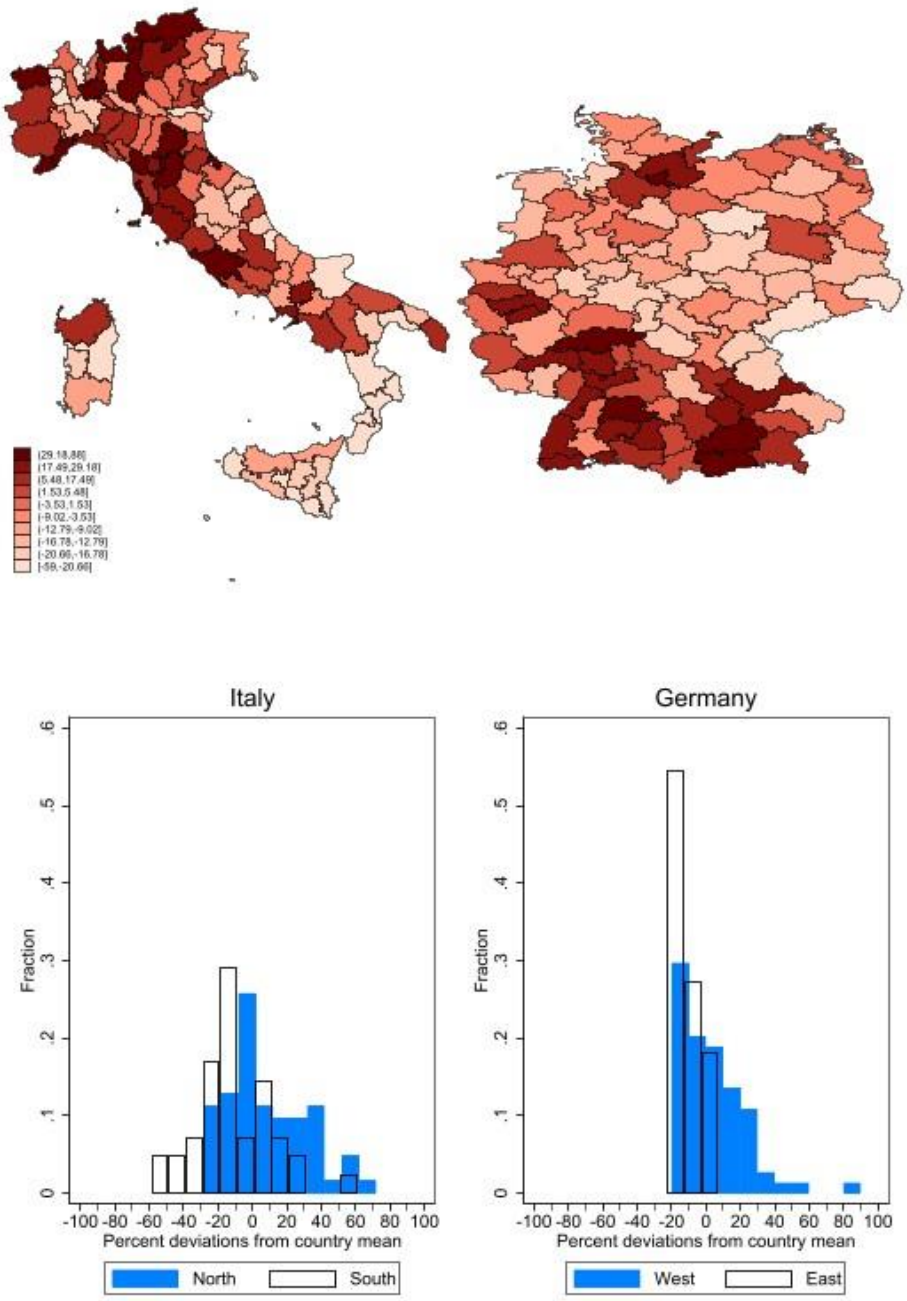
[22] To extend the analysis to the overall cost of living, we created the local Consumer Price Index (Local CPI) following the methodology proposed by Moretti (2013) and using data on regional measures of CPI from the Italian and German statistical offices (see Appendix B).

[23] Our theoretical model considers only housing prices and not the overall cost of living. Our considerations remain valid because differences in cost of housing represent the main source of geographical differences in local cost of living.

[24] To extend the analysis to the overall cost of living, we created the local Consumer Price Index (Local CPI) following the methodology proposed by Moretti (2013) and using data on regional measures of CPI from the Italian and German statistical offices (see Appendix B).

and west compared to the south and east, and that in Germany, the difference between west and east is more pronounced than Italy[25].

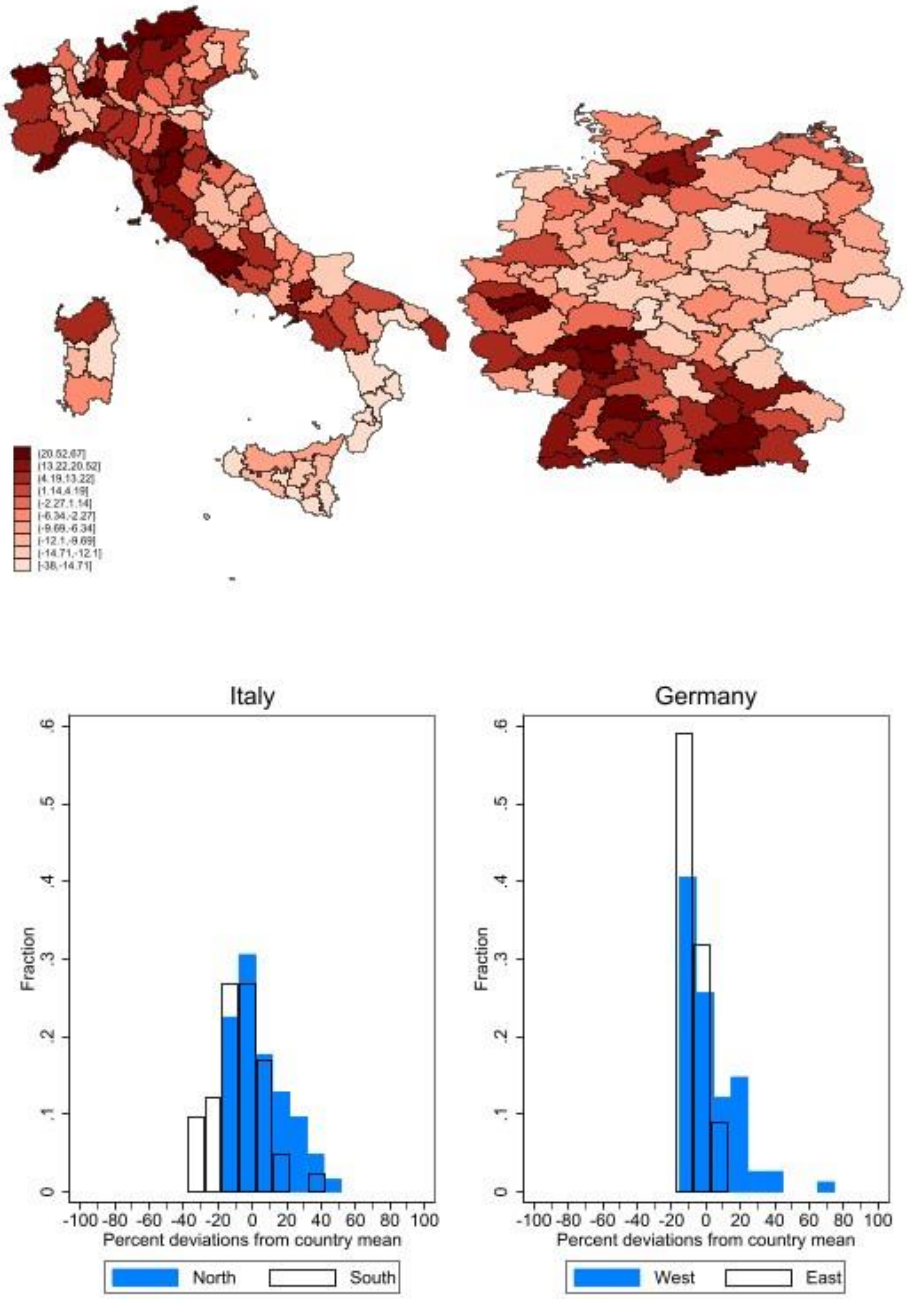
Figure 7. Housing costs distribution.



Note: This figure plots deviations from the country mean of the housing price index of each provinces in 2010. Means are not weighted by population. Source: Based on data from Appendix of Boeri et al. (2021, p.12).

[25] Our theoretical model considers only housing prices and not the overall cost of living. Our considerations remain valid because differences in cost of housing represent the main source of geographical differences in local cost of living.

Figure 8. Local cost of living index distribution.

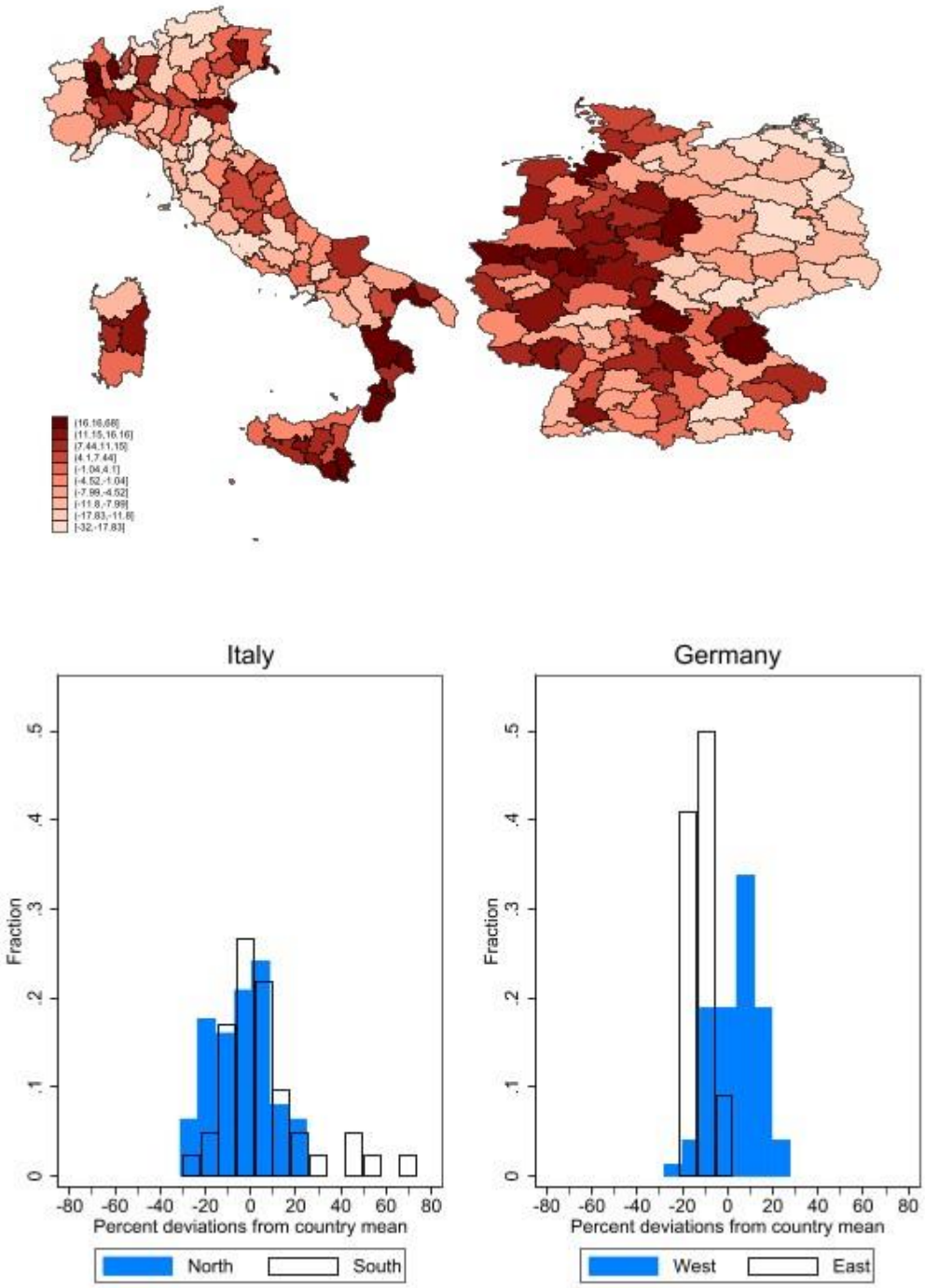


Note: This figure plots deviations from the country mean of the consumer price index (CPI) of each province in 2010. Means are not weighted by population. Source: Based on data from Appendix of Boeri et al. (2021, p.13).

Nominal wages are significantly higher in western than in eastern Germany, as are the price levels. In Italy, nominal wages are quite homogeneous, while price level in the north is higher than in the south. Figure 9 shows the real wage geographic distribution of both countries. It emerges a significant difference between Italy and Germany. In Germany, the west clearly has higher real wages than the east. In Italy, there are no clear

difference between the north and south in term of real wages, since the distributions of northern and southern Italy are almost completely overlapping. There is not a significant difference between the north and south in real wages, in contrast to Germany.

Figure 9. Real wages distribution.



Note: Derivation from the country mean of the mean of conditional real wages in euros of each province in 2010. Means are not weighted by population. Source: Based on data from "Wage equalization and regional misallocation: evidence from Italian and German provinces", Boeri et al. (2021).

Table 6 quantifies the differences between real and nominal wages across all macro-regions. Columns (1) and (3) show the nominal wage difference between North and South and between West and East respectively. The real wage difference between North and South in Italy is only 9.2%, whereas the West-East difference in Germany is twice as high, at 17.7%. The difference between the two countries in terms of nominal wage[26] disparity is even more pronounced: the West-East difference in Germany is 29.7%, while it is seven times smaller than in Italy with 4.3%. The difference in real wages is negative in Italy, because the southern Italian provinces are characterised by low living prices. Consequently, since nominal wages are similar between the northern and southern provinces, real wages in the south are higher. Analysing Figure 10, the blue dots (representing the provinces of western Germany and northern Italy) are almost all further to the right than the orange dots (the provinces east and south). This is because they are the most productive provinces and therefore have a higher added value than the provinces in the low productivity regions. If there were a net positive correlation between real wage and value added, the more productive provinces (in blue) would be in the upper right and the less productive ones (in orange) in the lower left. Therefore, in Italy, the elasticity of nominal wages with respect to value added is negative, while, in Germany, there is a slight positive correlation.

Table 6. Average wage internal differences between macro regions in Italy and Germany.

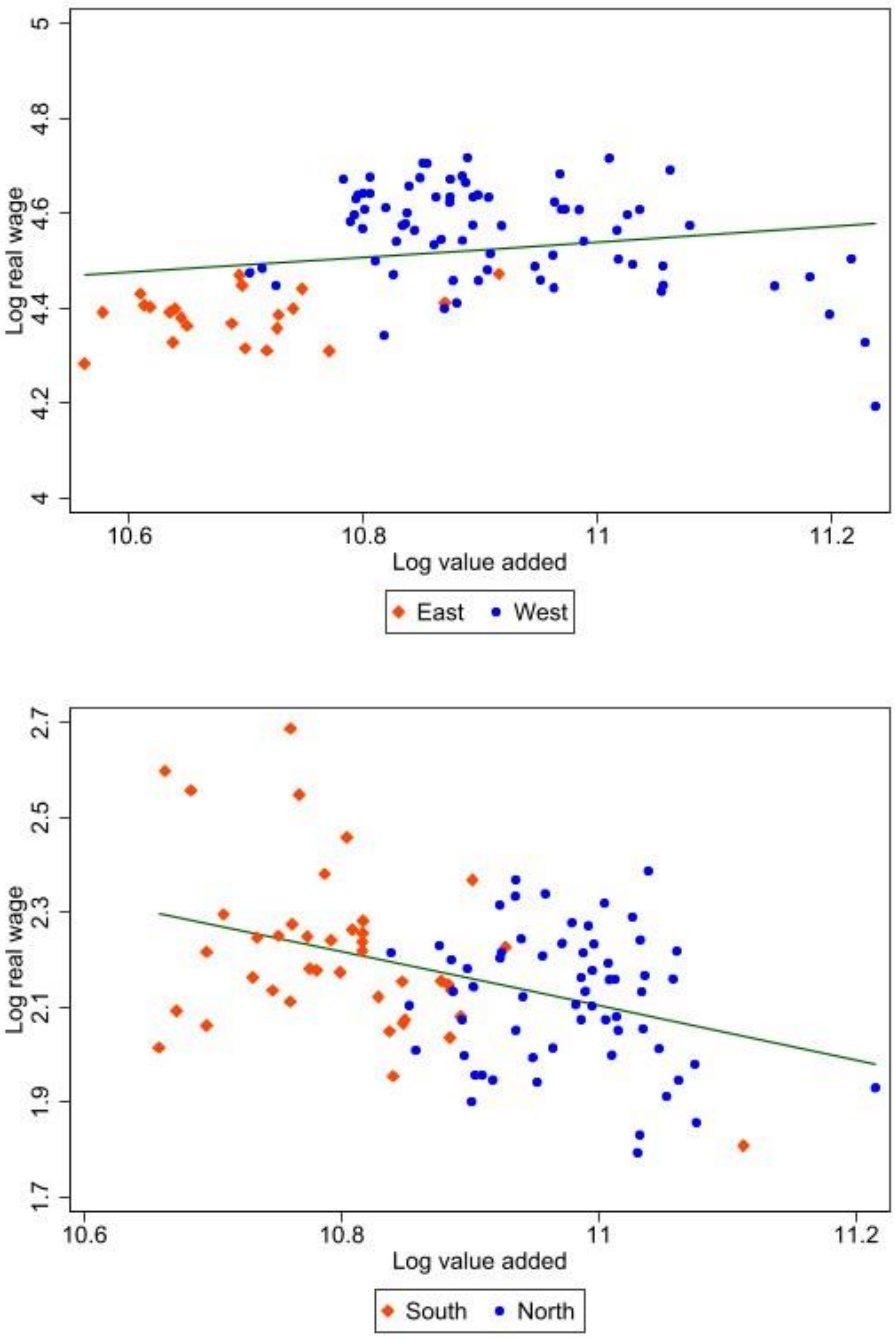
	North-South in Italy		West-East in Germany	
	(1)	(2)	(3)	(4)
	Nominal	Real	Nominal	Real
% Difference	0.0425	-0.0921	0.297	0.177
	(0.003)	(0.017)	(0.003)	(0.007)
Provinces:	103	103	96	96

Note: Entries in this table are the coefficients of the regressions of the difference in average wages (both nominal and real) between the macro-regions of Italy and Germany (North/South, West/East). For nominal wages we consider the years 2000-2014 for Germany and 2009-2013 for Italy, while for real wages we consider the years 2004-2014 (Germany) and 2009-2011 (Italy). All specifications control for year fixed effects and all standard errors in parentheses. Source: Based on data from Boeri et al. (2021, p.31).

[26] Wages we use for Italy are net of taxes. Given the progressive tax rates, taxes might compress wages. To correct for this, we use the gross and net wage data of all full-time workers from social security records (INPS) (since 2015) to generate a net-to-gross ratio for each province.

In Italy, less productive provinces have higher real wages, in Germany the opposite. There is no clear gap in real wages between north and south, but there is a significant difference in employment. The result is a spatial misallocation.

Figure 10. Real wage and value added



Note: This figure shows the relationship between log mean conditional real wages and log mean value added across provinces in 2010. Each province in represented by a dot. Blue dotes are the most productive provinces. Germany is represented above and Italy below. Source: Based on data from Boeri et al. (2021, p.32).

5. Counterfactual analysis

This chapter shows the counterfactual calculation to quantitatively estimate the costs of regional misallocation in Italy. The main issue with the Italian wage-setting system is that it causes a misalignment between wages (both nominal and real) and productivity. As seen in Table 4, which shows the elasticity between nominal wages and added value, Italy has an elasticity of 0.195, while Germany has an elasticity of 0.739:

$$\eta_{nw,va}^{ger} = 0.739 > 0.195 = \eta_{nw,va}^{ita}$$

This means that the German system is better able to align nominal wages with productivity. In fact, in Germany, wages adapt better to the local productivity of each region, consequently producing and hiring more. Namely, there is greater efficiency then Italy.

Elasticity of the non-employment rate relative to added value is lower in Italy than in Germany. Italian less productive regions, that are in the south, have higher non-employment rates, whereas in Germany, this is not the case.

To quantitatively estimate the costs of regional misallocation in Italy, it is necessary to conduct a counterfactual analysis. Considering a scenario in which Italy has implemented reforms aimed at making the mechanism more flexible (Counterfactual I), with the goal of better linking nominal wages to productivity to the same extent as in Germany, Italy's elasticity would be as high as Germany's: $\eta_{ita}^{\prime w} = 0.739$.

Similarly, regarding the elasticity of the non-employment rate[27] with respect to added value, the value of Italian counterfactual elasticity would be the same German one in status quo: $\eta_{ita}^{\prime e} = 0.252$.

The objective is to calculate how economic outputs would change in this scenario. In the North, there are no changes because we assume that there is full employment. What changes are the nominal wages, the employment rate, and the aggregate income in the south (and consequently, the national values also change). In this scenario, one finds all the counterfactual nominal wages and employment rates for all the provinces in the South

[27] In the code Boeri seems to have used the elasticity of the employment rate with respect to value added, which is 0.1185 (Appendix C).

and then computes a weighted average based on the working-age population[28] to determine the counterfactual average wage and the counterfactual average employment rate in the south. After that, one calculates the national averages to understand the impact of the wage-setting system on macroeconomic outputs (wages, employment, labour income). To perform these calculations, the following two reference formulas are used:

$$w_c' = \left[1 - \eta_{ita}^{'w} \frac{\overline{A_n} - A_s}{\overline{A_n}} \right] * \bar{w}_n \tag{14}$$

$$E_c' = \left[1 - \eta_{ita}^{'e} \frac{\overline{A_n} - A_s}{\overline{A_n}} \right] * \bar{E}_n \tag{15}$$

where w_c' and E_c' are counterfactual nominal wage, employment rate for each southern province; A_s is the value added for each southern province; $\bar{A}_n$, $\bar{w}_n$ and $\bar{E}_n$ are value added, wage and employment rate of the median province in the North; and $\eta_{ita}^{'w}$ and $\eta_{ita}^{'e}$ are the elasticities of nominal wages and the employment rate with respect to added value[29]. Notice that wages, employment, or productivity in Italy are not considered the same as in Germany; only the elasticity of nominal wages or the employment rate with respect to added value is assumed the same of German in status quo.

At the end of the analysis, the counterfactual wages of the provinces in the South are lower than the wages in the status quo (i.e., the actual Italian scenario), while the counterfactual employment rates are much higher than in the status quo. If the demand for labour is elastic (which is likely in a small open economy, like a province), the increase in employment is proportionally greater than the decrease in wages, resulting in a positive effect on total labour income. Therefore, those who would lose out are the people who are already working in the South since their counterfactual wages would be lower. On the other hand, unemployed people in the South would benefit, as it would be easier to find a job. Overall, the effect is positive, as the counterfactual labour income would be greater than that of the status quo.

[28] Since Italian provinces have different working age population, it would not be fair to consider provinces with completely different working age populations with the same weight. By weighing the averages, we have a more accurate mosaic of Italy's wage and employment levels (calculations in Appendix C).

[29] The median values for northern Italy are 59053.5 (value added), 8.653 (nominal wage) and 71.23 (employment rate).

Considering another scenario (Counterfactual II), in which the Italian system manages to perfectly align wages with added value, the elasticity of wages with respect to added value would be: $\eta''^{w}_{ita} = 1$

In Counterfactual II, regarding employment, the counterfactual labour market balances at: $E''_c = E_n$. This is because if wages were to fully adjust to the added value, employment would not adjust at all to the added value, and thus the elasticity of employment with respect to the added value would be zero: $\eta''^{e}_{ita} = 0$. Applying equation 14, if the elasticity is zero, $E''_c = E_n$. Meanwhile, the reference formula for counterfactual wages changes in the following way:

$$w''_c = \left[1 - \frac{\overline{A_n} - A_s}{\overline{A_n}} \right] * \bar{w}_n \tag{16}$$

In this scenario, the same changes occur as in Counterfactual I. The counterfactual wages of the provinces in the South are lower than the wages in the status quo, while the counterfactual employment rates are much higher than status quo's values, and, finally, that the counterfactual labour income is higher than in the status quo. What changes is the quantitative measure and the impact of these variations.

Table 7 shows the results of the entire counterfactual analysis[30]. It shows that the counterfactual wages in the South (and consequently in Italy) are lower than in the status quo. On the other hand, the counterfactual employment rates in the South (and consequently in Italy) are much higher than in the status quo. It is important to note that the positive change in employment is proportionally greater than the negative change in the hourly wage. Therefore, as the counterfactual aggregate income in the South (and in Italy) is higher than in the status quo, the effect of the German elasticities on Italian economic variable outcomes is positive.

[30] I independently went through all the calculations made in the Boeri et al. paper (2021) in an excel file. I went through the whole of the State code (within the limits of my competence) in order to understand all the steps of the counterfactual calculation. In Appendix C, there are all the steps performed on excel to arrive at Table C2, which is the result of all the calculations performed by me, corresponding to table 7, which is shown with in the Boeri's paper. The substantial differences are in the values of wages in the north and in the counterfactual I employment rate in the south. All other discrepancies (in deltas and percentages, for example) are simply a consequence of the development of the calculations with a different wage in the north and counterfactual I employment rate in the south. These discrepancies in values are however very small and fall within the scale of 2/3 percentage points. Therefore, the values I found, do not change the substance of the implications made so far and the conclusion of the paper remain unaltered. In Appendix C, I explain my calculations in depth and why the results are slightly different from those in the paper.

Table 7. Counterfactual exercise for Italy.

	South		North		Italy		
	(1) Level	(2) Change	(3) Level	(4) Change	(5) Level	(6) Change	(7) %Change
Average hourly wage *in euros*							
Status quo	8.36		8.68		8.54		
Counterfactual I	7.84	-0.52	8.68	0.00	8.31	-0.23	-2.60
Counterfactual II	7.56	-0.80	8.68	0.00	8.18	-0.36	-4.13
Employment rate: *in % corrected for informal work*							
Status quo	57.32		71.00		64.86		
Counterfactual I	70.17	12.85	71.00	0.00	70.63	5.77	11.04
Counterfactual II	71.24	13.92	71.00	0.00	71.11	6.25	11.95
Aggregate labour income per capita: *in euros per month*							
Status quo	766.63		986.68		887.89		
Counterfactual I	881.93	115.30	986.68	0.00	939.65	51.76	7.51
Counterfactual II	861.44	94.81	986.68	0.00	930.46	42.56	6.22

Note: In each of the three panels (for wages, employment and labour income), the first row represents the status quo in 2010, the second row the first counterfactual and the third row the second counterfactual. The items in column (3) are the averages of wages, employment and income in the north. column (1) are the weighted averages for the working-age population of wages, employment rates and income for each province in the south. Column (5) is the weighted averages over the working age population for Italy as a whole. While the entries in columns (2), (4) and (6) represent the average changes corresponding to each counterfactual scenario. Column (7) shows the average percentage change in all Italian provinces, corresponding to each counterfactual scenario. Income is the product of hourly wages, employment rate and hours worked (160 hours per month). Source: Based on calculation from Boeri et al. (2021, p.39).

5.1 Average hourly wage

This subchapter goes into the details of the calculations performed to find the values in the first panel on average hourly wage of Table 7. The value of the elasticity of nominal wage with respect to value added is $\eta_{ita}^{'w} = 0.739$. All the value added for each province in the South are given in the data analysis[31]. For the calculation of the values of the average hourly wage counterfactual I, the values of the median northern province for value added and nominal wages, and the value of the elasticity were used. While for counterfactual II, the elasticity is considered equal to 1, since a total adjustment of wages to value added is assumed. Table 8 presents the values used to arrive at the results of the first panel.

[31] Value-added is computed across all industries in each geographic area and it is divided by employment in the corresponding area (Appendix C).

Table 8. Values for counterfactual calculations for Average hourly wage

North Italy	Values
Median value added per worker ($\bar{\boldsymbol{A}}_{\boldsymbol{n}}$)	59053.5
Median nominal wage ($\bar{\boldsymbol{w}}_{\boldsymbol{n}}$)	8.653
Elasticity of nominal wage on value added ($\boldsymbol{\eta}'^{\boldsymbol{w}}_{\boldsymbol{ita}}$)	0.739

Note: These values come from quarterly statistics data of ISTAT (Istituto Nazionale di Statistica). Information is only available from 2009, and we focus on the period 2009-2013. Elasticity is only used for Counterfactual I, in Counterfactual II the elasticity is equal to one. Source: Based on data from Boeri et al. (2021).

The nominal wage in the status quo for Southern Italy is 8.36, in Counterfactual I it is 7.84, and in Counterfactual II it is 7.56. Intuitively, the value in Counterfactual II is lower than in Counterfactual I because nominal wages adjust perfectly to productivity, allowing businesses in all regions to pay wages at the level of the local market price.

Since nominal wages in the North do not change, the counterfactual wages for Italy are lower than in the status quo. Table 7 shows that the average nominal hourly wage for Italy is 8.54, while it is 8.31 for Counterfactual I and 8.18 for Counterfactual II. There is a decrease of -0.23 for Counterfactual I and -0.36 euros for Counterfactual II, which, in percentage terms, would be -2.60% and -4.13%, respectively.

5.2 Employment rate

In order to calculate of the second panel values of corrected employment rate in counterfactual I, one uses the elasticity of non-employment rate with respect to value added $\eta'^{e}_{ita} = 0.252$, the values of the median northern province for value added and employment rate. While for counterfactual II, the elasticity is considered equal to 0, since a total adjustment of wages to value added is assumed. Consequently, from Equation 14, $E_c = E_n$. Table 9 presents the values used to arrive at the results of the first panel. The employment rate in the status quo for Southern Italy is 57.32, in Counterfactual I it is 70.17, and in Counterfactual II it is 71.00. Therefore, there is a significant increase in the employment rate in both Counterfactual I and II compared to the status quo, with an increase of 12.85 and 13.92 percentage points in the south.

Table 9. Values for counterfactual calculations for Employment rate.

North Italy	Values
Median value added per worker ($\bar{A}_n$)	59053.5
Median employment rate ($\bar{E}_n$)	71.23
Elasticity of employment rate on value added (η'^{e}_{ita})	0.252

Note: These values come from quarterly statistics data of ISTAT (Istituto Nazionale di Statistica). Information is only available from 2009, and we focus on the period 2009-2013. Elasticity is only used for Counterfactual I, in Counterfactual II the elasticity is equal to one. Source: Based on data from Boeri et al (2021).

Since the employment rates in the North do not change, the counterfactual employment rates for Italy are globally higher than in the status quo. In fact, the employment rate for Italy is 64.86 in the status quo, while it is 70.63 for Counterfactual I. There is an increase of 5.77 points, which equals 11.04% for Counterfactual I. As for Counterfactual II, there is an increase of 13.92 points in the South and 6.25 points for Italy, equivalent to 11.95%.

5.3 Aggregate labour income per capita

The aggregate labour income is the product of the average monthly earnings (i.e. hourly wage times 160 working hours per month) and the employment rate. One uses this formula to calculate the aggregate labour income of southern and northern Italy, and the whole of Italy in the status quo and in the two counterfactuals:

$$Y_r^{inc} = w_c * E_c * 160$$

Table 7 shows the results of these calculations in the last three rows. Southern Italy has an aggregate labour income per capita of €766.63 per capita in the status quo, while in the two counterfactuals, it increases by €115.30 and €94.81 per capita, respectively, reaching €881.93 and €861.44 per capita. In northern Italy, the aggregate labour income per capita is €986.68 per capita and remains the same in all three scenarios. Overall, Italy has a value of €887.89 per capita in the status quo, €939.65 per capita in Counterfactual I, and €930.46 per capita in Counterfactual II, increasing by €51.76 and €42.56 per capita in the two counterfactuals, respectively, which translates to an increase of 7.51% in the first counterfactual and 6.22% in the second.

These results show that, although there is a decrease in nominal hourly wages in the south, this decrease is proportionally smaller than the increase in employment in the counterfactuals. Consequently, there is a positive effect on aggregate labour income. To summarize, if Italy were to implement changes in its wage-setting system to achieve the same elasticity of nominal wage with respect to value added as the German system, namely if it could relate wages with productivity as Germany there would be a significant positive effect on aggregate labour income.

Additionally, it is interesting the comparison between the two counterfactuals. Regarding wages, the first counterfactual has a higher nominal wage than the second, while, in terms of employment, the second has a higher employment rate than the first. Concerning the aggregate labour income per capita, Counterfactual I has a higher value than Counterfactual II. Thus, the difference in aggregate labour income between the two counterfactuals is €20.49 per capita (881.93 - 861.44). In other words, in Counterfactual I, there is a greater effect on income than in the second. Therefore, for Italy, there would be a greater impact on income if the elasticity was aligned with the German one, rather than achieving complete wage adjustments to productivity elasticities.

It is necessary to point out that if Italy were to adopt the same measures as Germany, it is not certain that the Italian elasticity would automatically assume the values of the German elasticities. This is because the country's production structure is profoundly different. Italy is characterised by a dense network of small and medium-sized enterprises[32], whereas in Germany, big companies (technology and automobile companies) dominate. Therefore, the reception of new reforms and the ease with which they can be implemented may be different between Italy and Germany. This research supports that if Italy were to adopt measures capable of bringing this elasticity up to German levels, there would be considerable benefits, although these measures or reforms may differ according to the economic structure of a country. The “marketisation” of wage bargaining would strongly limit regional misallocation, which imposes substantial income costs on Italy.

[32] Italy is a country characterised by small companies compared to other countries. In fact, in Italy companies with less than 100 employees account for almost 70% of the total. Italian companies have an average of 4.4 employees, while German companies have on average, 10.3 employees (Guiso 2003).

6. Discussion

Last chapter quantifies the aggregate costs for Italy through counterfactual calculations that are 11.04% in employment rate and 7.51% of aggregate labour income. Italy could significantly increase employment in the south by making its wage setting mechanism more flexible, allowing wages to adjust to local market levels. To arrive at this important implication, it was necessary to carefully study the characteristics of the systems of Italy and Germany, devise a theoretical model, analyse the empirical evidence and carry out counterfactual exercises. Throughout this process, however, no consideration was given to local amenities, which are an important element that could expand on the implications found so far. This chapter presents a discussion on the research findings in a discussive manner to demonstrate the study's validity and rigor, even though some aspects were not addressed.

For example, the analysis performed considered the following utility function in the theoretical framework:

$$\Omega_r = \frac{w_r}{p_r^{\sigma}}(1 - u_r)^{\delta}$$

where w wage, $u_r = 1 - (E_r \backslash L_r)$ unemployment rate, p_r price level, σ indicates the sensitivity of utility to price changes and δ measures the elasticity of value with respect to the unemployment rate.

Utility function Ω_r does not include local amenities, which instead could be very influent. In equilibrium, a situation in which it is indifferent to a citizen in which macro region to live in, this function is equal between north and south. Since local amenities impact on a person's choice of location, it is appropriate that the utility function also includes this element even if measuring local services is notoriously very difficult. Location choices depend on a multitude of factors, including household and trade productivity, land availability, public goods and, above all, quality of life. The latter appears to be the most decisive factor in people's choice of location (Albouy, Christensen, and Sarmiento-Barbieri 2019). Location choices depend on two different types of forces: the forces of agglomeration, which concentrate economic activity in circumscribed urban areas, and the forces of dispersion, which, on the other hand, ensure an even distribution of population between urban and non-urban areas (Redding 2011, p. 24).

The goal of this chapter is not to incorporate welfare and amenities into the model but is to show that the considerations made in the previous chapters remain valid. When analysing empirical data, one should consider the existence of systematic geographic differences in local services that influence people's location choices, such as climate, crime rates, pollution, quality of local public goods, entertainment, and cultural activities. This could have a significant impact on the structure of labour force in both countries. Consequently, housing market prices would also be affected, as well as real wages. Some aspects of internal migration flows in Italy should also be clarified, as they alter the composition of workers in various regions of Italy.

It is necessary to better understand the reasons for the massive migration (especially of the youth) from the south to the north. Internal migration flows started to increase in 1996. In 1998, emigration flows from the south to the rest of Italy reached 129,000 units, a level not seen since 1974. During the period 1996–2005, there was a positive trend in interregional migration flows. In the decade from 2012 to 2021, the South lost 525,000 residents[33] (ISTAT 2021).

It should be noted that migrants heading north are much younger than those heading south. This significantly impacts the local distribution of the qualified and technologically prepared workforce. A more in-depth analysis of individual migration decisions indicates that young, highly educated families tend to move to places with a more productive economic environment. This trend is also most noticeable in highly educated couples, who have a better choice of where to work than couples with a low level of education. On the other hand, older couples and those close to retirement move from more productive areas to areas with high-value consumer services (Chen and Rosenthal 2008, p.1).

In fact, we could imagine that the presumed low quality of local amenities in the South could encourage people to migrate north. If real wages are similar, but the local amenities are much worse, the utility of living in the south would be lower. However, it is not the case. In other words, if low productivity areas had worse services real wages would have to be higher to reach equilibrium, i.e. indifference to migration. However, one cannot state this, because it is not clear whether local amenities are better in low-

[33] These data, collected by each municipality in their official civil registers, provide the number of Italian citizens who, during a year, cancelled their residence in their region to move to another. About 1,138,000 people moved from the south and Islands to the centre-north, while about 613,000 moved in the opposite direction.

productivity regions or in high-productivity ones. Table 10 quantifies the correlation between local services and local value added.

Table 10. Local amenities and Value added

Index of pleasant climate (2019)	-0.24	-0.17
	(0.05)	(0.07)
Temperature excursion over the year (2018)	0.17	0.17
	(0.04)	(0.05)
Air pollution—PM10 in the air (2019)	0.20	0.22
	(0.08)	(0.12)
Population density (inhabitants per km2) (2017)	0.78	1.28
	(0.27)	(0.39)
Number of crimes per 100 thousand inhabitants (2019)	0.34	0.34
	(0.08)	(0.13)
Number of murders per 100 thousand inhabitants (1999)	-1.84	-1.54
	(0.21)	(0.31)
Number of students per class (primary, 2017-2018)	0.16	0.19
	(0.03)	(0.05)
Number of students per class (junior high school, 2017–2018)	0.16	0.15
	(0.02)	(0.03)
Number of students per class (high school, 2017–2018)	0.11	0.12
	(0.02)	(0.03)
Quality of health system (health related emigration, 2019)	-0.93	-1.19
	(0.20)	(0.30)
GP doctors per 1,000 inhabitants (2019)	-0.23	-0.21
	(0.06)	(0.08)
Number of cultural shows per 10 square km (2019)	1.79	2.06
	(0.32)	(0.47)
Ranking of local food and wine quality (2007)	1.49	0.57
	(0.40)	(0.58)
Region FE:	No	Yes

The entries indicate the coefficients of a regression of the logarithm of a disamenity indicator on the logarithm of the value added of the 103 provinces. In Column (2) the regression includes fixed effects for the North (standard errors in brackets). The disamenity indicators and the climate index come from the 'Quality of Life Survey' conducted annually by the Italian business newspaper 'Sole24Ore' (https://lab24.ilsole24ore.com/qualita-della-vita-2019/index.php). The annual temperature range represents the average difference between daytime and night-time temperatures in degrees Celsius. Air pollution is measured as the average concentration of PM10 particles per cubic metre of air in 2019. The number of students per class is provided by the Ministry of Education. The quality of the healthcare system is determined by the percentage of hospitalised patients who were treated outside the region. The ranking of the quality of food and wine is calculated by Il Sole 24 Ore based on data provided by CENSIS. Source: Based on data from Boeri et al. (2021, p.33).

Table 10 shows the correlation between different variables related to quality of life and provincial logarithmic added value. The first three rows focus on climate quality, temperature variations, and pollution. Good weather is negatively and significantly correlated with added value across provinces, while temperature variations (which are considered a disadvantage) and pollution are positively correlated with added value. The next three rows display population density, number of crimes, and the number of homicides. It appears that regions with high added value have higher population density

and a higher total number of crimes, but a significantly lower number of homicides. Following that, there are indicators of school quality that show a positive correlation between these indicators and added value. On average, a class in the south has one student less compared to the north across all three stages of education. Next, there are two rows regarding the quality of the healthcare system, indicating that the fraction of citizens seeking care outside their region of residence is significantly higher in areas with low added value. However, there are 99 general practitioners per 1000 inhabitants in the South and 89 in the North. In fact, there is a negative correlation between the number of doctors per 1000 inhabitants and added value. The last two items provide evidence on cultural services and the quality of food in restaurants. Both indicators are positively correlated with added value, although they are likely endogenous.

In summary, in the high-productivity provinces of the North, less favourable weather, higher levels of pollution and congestion, and more property crimes are observed compared to the South, but schools tend to have larger classes. On the other hand, the less productive provinces in the South enjoy better weather, cleaner air, and less congestion, but with lower quality public services, including healthcare. Therefore, it is not at all clear where a better level of local amenities exists. At this point the question is why so many people choose to move to the North, even if there are no obvious differences in real wages and local amenities between the North and South. A possible explanation lies in the expected wages of workers. In fact, migrants should consider the expected wage, which depends on the probability of finding a job, estimated through the unemployment rate, in addition to the level of real wages.

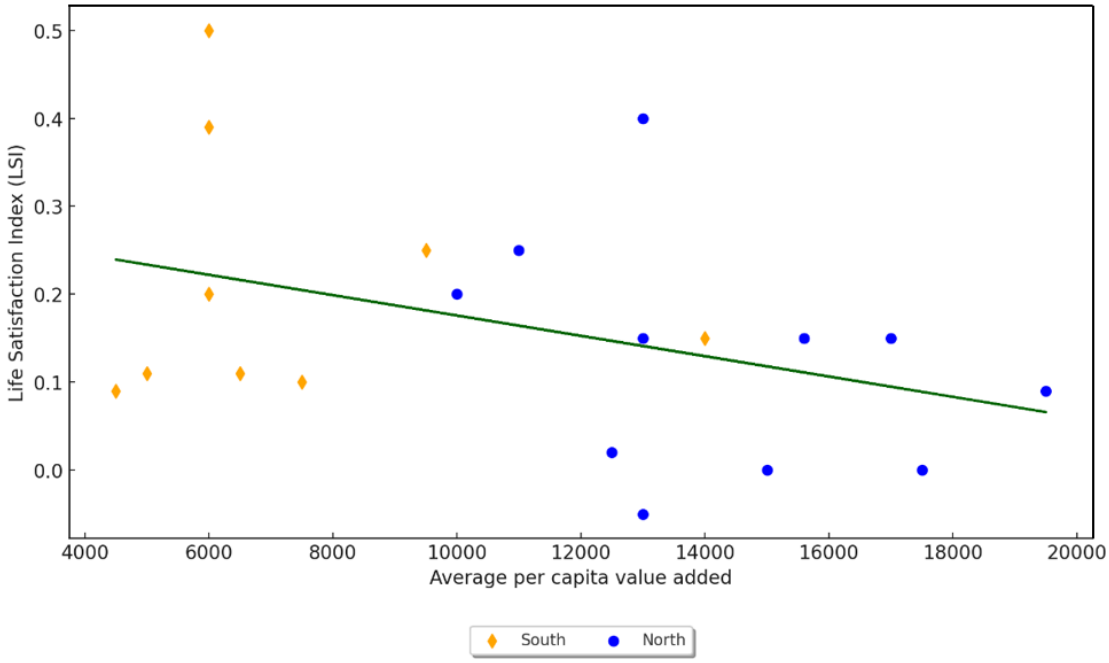
If differences in employment rates decrease, migration becomes a less attractive option for increasing expected net income. This is true in Italy because there is not so much difference between Northern and Southern wages. In fact, in this case, the probability of finding a job impacts much more on the expected wage than does the level of wages (Etzo 2011, p.3). In other words, if there is a very high non-employment rate, the probability of finding a job decreases, and consequently, the expected wage of workers in the south is much lower than that of workers in the north. This is why massive migration from the south to the north is understandable, despite there being no clear difference in terms of real wages and local amenities between the North and South.

Separate evidence that could help understand geographic differences in services is represented by the location choices of public sector employees. Public employees often prefer to move from the North to the South, where nominal wages are the same, but the

cost of living and other conditions are perceived as better. So, this means that if a worker has the certainty of a job contract, they are more likely to prefer the south over the north. The public sector offers greater stability than the private sector. Therefore, the expected value of a worker aiming for a public sector position may be higher than that of another aiming for the private sector in southern Italy, precisely because the probability of finding a stable job in the public sector is higher. This is why many public employees want to move or to return to southern Italy.

Figure 11, from the analysis of determinants of life satisfaction by ISTAT, shows that there is no positive correlation between the Life Satisfaction Index and value added, i.e. productivity, when controlling for the regional non-employment rate. Therefore, differences in life satisfaction between regions seem to depend more on employment status than on productivity, suggesting that well-being is not strictly linked to regional economic productivity (ISTAT 2019, p.7).

Figure 11. Relationship between life Satisfaction Index and value added.



Source: Istat 2019, Analysis of determinants of life satisfaction (p.6), Eu-silc survey.

A high non-employment rate significantly decreases the likelihood of being satisfied. Living in regions with high unemployment rate is a risk factor for satisfaction. In synthesis, for an average person from southern Italy, finding a job has more impact on life satisfaction than the level of salary. The amount of salary is certainly important in life satisfaction, but employment status is even more important for people in the south (ISTAT 2019, p.6).

The role of institutions is crucial in this. Since they affect the output of economic variables, such as the level of wages and the employment rate, they also have an impact on the labour force's mobility across enterprises (Marsden 1990, p.1). Obviously, it is very difficult to address these aspects that determine people's life choices. The aim of this chapter was to show that the analysis conducted in previous chapters does not change, even when considering broader aspects such as local amenities and, consequently, internal migration within the country.

A horizon for further future research could be the introduction of the state, local amenities and all those factors that impact on labour mobility into the model, so as to broaden and deepen it. Our analysis does not aim to integrate all these elements into the model, but we have attempted to empirically study the role of local amenities and welfare in the dynamics between the wage setting system and the economic variables considered.

This topic deserves the attention from the literature because it has a significant impact on the development on the countries' economy and society, and if this issue is addressed properly, the solutions to change the current situation can be found.

7. Conclusion

This analysis studies the local and aggregate effect of different wage setting systems on wages, non-employment rates, housing costs, and real wages by comparing the Italian and German spatial distribution. Italy and Germany face similar productivity geographical distribution, but they have two opposite wage bargaining model. In fact, Italian system is centralized and rigid, while German is decentralised and more flexible.

It was necessary to idealise a model considering two regions with different productivity, assuming zero mobility costs, no heterogeneous preferences for location, fixed population of each region, upward sloping housing supply and identical regions apart from their level of TFP. The model was developed in two different scenarios: free-market and fixed-wage scenario. In free-market scenario (extremization of German system), employment rate, capital, nominal wage and housing price are higher in the more productive region.

In fixed-wage case (extremization of Italian system), employment rate, capital and housing price are higher in the more productive region, but the real salaries are higher in the less productive (due to low prices). In addition, there is an important regional gap in employment rate in fixed-wage scenario. In brief, wage inequality is more accentuated in free-market scenario and employment inequality is stronger in fixed-wage case. Another important implication of the model is that labour income in the free-market scenario is higher than in the fixed-wage case. Thus, there is regional misallocation in fixed-wage case, i.e. inefficiency in the labour market.

The model is in line with the empirical data, taking data, principally, from Institute for Employment Research (IAB) for Germany and National Italian Statistical Institute (ISTAT). The empirical evidence shows that Italian and German value-added regional distributions are similar. Then, for Italy, there is not a remarkable difference between North and South in nominal and real wages, but there is a huge difference in employment rate.

On the other hand, in Germany there is a sharp difference in nominal and real wage between east and west, but there is no significant difference in employment rate. The contrast between the two countries is that German productivity is correlated with real wages, whereas the Italian is not. In Italy, the correlation is not clear. In fact, Germany suffers a huge wage inequality, Italy struggle with an important employment inequality. This empirical evidence determine that the model is consistent with the reality.

Furthermore, through the counterfactual analysis, we show that if Italy would have the German value of elasticity between mean nominal wages on mean value added, it would have a higher aggregate labour income. This means that Italian system is inefficient and provoke regional misallocation. Counterfactuals exercise quantifies this losses and, if Italy would have the same German elasticities (of wage and employment rate with respect to value added), aggregate employment and earnings would increase by 11.04% and 7.51%, respectively.

The model is clear and simple and describes the dynamics of the wage setting mechanism on the economic outputs (wages and employment). However, the model does not consider the impact of local amenities. The implications of the model are still consistent with the data findings, but future studies can implement the model with local amenities. Completing the model would help to accurately describe this phenomenon, because, since qualitative implications in models of economic geography very often change abruptly when the quantitative equilibrium goes beyond critical limits, the more precise and complete the model, the more we will be able to understand the phenomenon in greater depth (Krugman 2008, p.5).

Geographical economic models are always delicate to deal with, because each region has its own peculiarities. One would have to have a plethora of elements at one's disposal to paint a complete picture.

In conclusion, this study states that a centralised and rigid wage setting system mitigate wage inequality, but it sharpens Regional Misallocation, especially in countries with big internal differences in productivity, because wages do not fit with productivity. Therefore, our analysis supports that Italian regional misallocation come from the rigidity of the wage setting system. This is not to say that Italy should adopt the same measures that Germany has adopted since unification to decrease allocative inefficiency.

This analysis supports that it would be opportune for Italy to adopt measures to make the wage setting system more flexible with the objective of linking wages more closely to productivity (i.e. increasing the elasticity of wages to value added). The way Italy should achieve this flexibility is not the goal of this research, but it points out that a decentralised system that could adapt the level of wages according to the local labour markets of each region would be suitable for the country.

References

Afonso, A., Dorigatti, L., Molina, O., & Tassinari, A. (2021): Labour market (de)regulation and wage-setting institutions in Mediterranean capitalism, in: *Mediterranean Capitalism Revisited*, chapter 5, pp. 115-148.

Albouy, David, Peter Christensen, and Ignacio Sarmiento-Barbieri (2019): Unlocking amenities: Estimating public good complementarity, *Journal of Public Economics*, vol. 182, pp. 104-110.

Boeri, Tito (2014): Two-tier bargaining, *SSRN*, Paper No. 8358.

Boeri, Tito, et al. (2021): Wage equalization and regional misallocation: evidence from Italian and German provinces, *Journal of the European Economic Association*, 19.6, pp. 3249-3292.

Brändle, Tobias (2017): Flexible collective bargaining agreements: Still a moderating effect on works council behaviour?, *Managerial and Decision Economics*, 2017, vol. 38.8, pp. 1137-1153.

Chen, Yong, and Stuart S. Rosenthal (2008): Local amenities and life-cycle migration: Do people move for jobs or fun?, *Journal of Urban Economics*, 2008, vol. 64.3, pp. 519-537.

Devicienti, Francesco, Bernardo Fanfani, and Agata Maida (2019): Collective bargaining and the evolution of wage inequality in Italy, *British Journal of Industrial Relations*, vol. 57.2, pp. 377-407.

Erbetta Fabrizio, Petraglia Carmelo (2011): Drivers of Regional Efficiency Differentials in Italy: Technical Inefficiency or Allocative Distortions?, *Growth and Change*, 2011, Vol. 42 No. 3, pp. 351–375.

Etzo, Ivan (2011): The determinants of the recent interregional migration flows in Italy: A panel data analysis, *Journal of regional science*, vol. 51.5, pp. 948-966.

Faini, Riccardo (1999): Trade unions and regional development, *European Economic Review*, vol. 43.2: 457-474.

Flanagan, Robert J. (1999): Macroeconomic performance and collective bargaining: an international perspective, 1999, *Journal of economic literature*, vol. 37(3), pp. 1150-1175.

Gnocato Nicolò, Modena Francesca, Tomasi Chiara (2020): Labour market reforms and allocative efficiency in Italy, *Labour Economics*, 2020, vol. 67, 101938.

Guiso, Luigi (2003): Small business finance in Italy, *EIB papers*, 2003, vol. 8.2, pp. 121-149.

Hornbeck, Richard, and Enrico Moretti (2018): Who benefits from productivity growth? Direct and indirect effects of local TFP growth on wages, rents, and inequality, *National Bureau of Economic Research*, 2018, No. 24661.

ISTAT (2019). *Analysis of determinants of life satisfaction.*

ISTAT (2021). *Report migrazioni 2021*

Iuzzolino, Giovanni, Guido Pellegrini, and Gianfranco Viesti (2013): Regional Convergence, in: Toniolo, Gianni: *The Oxford Handbook of the Italian Economy Since Unification*, 1. Edition, Oxford University Press, pp. 571–598.

Lucifora, Claudio, and Daniele Checchi (2002): Unions and labour market institutions in Europe, *Economic Policy*, 2002, vol. 35, pp. 361-408.

Krugman, Paul (1998): What's new about the new economic geography?. *Oxford review of economic policy*, (1998), vol. 14(2), pp. 7-17.

Marginson, Paul & Welz, Christian (2015): European wage-setting mechanisms under pressure: Negotiated and unilateral change and the EU's economic governance regime, *European Review of Labour and Research*, 2015, vol. 21(4), pp. 429-450.

Marsden, David (1990): Institutions and labour mobility: occupational and internal labour markets in Britain, France, Italy and West Germany, *Palgrave Macmillan London UK*, 1990, (pp. 414-438).

Matysiak, Anna, Tomáš Sobotka, and Daniele Vignoli (2021): The Great Recession and fertility in Europe: A sub-national analysis, *European Journal of Population*, 2021, vol. 37.1, pp. 29-64.

Moretti, Enrico (2013): Real wage inequality, *American Economic Journal: Applied Economics*, 2013, vol. 5.1, pp. 65-103.

OECD (2017). "Collective Bargaining in a Changing World of Work." In *OECD Employment Outlook 2017*. OECD Publishing

OECD (2018). "The Role of Collective Bargaining Systems for Good Labour Market Performance." In *OECD Employment Outlook 2018*. OECD Publishing.

Ochel, Wolfgang (2005): Decentralizing wage bargaining in Germany—a way to increase employment?, *Labour*, 2005, vol. 19.1, pp. 91-121.

Poy, Samuele (2017): L'eredità storica nelle politiche: dalle gabbie salariali alla contrattazione decentrata, *Meridiana*, 2017, vol. 90, pp. 231-254.

Redding, Stephen J. (2011): Economic geography: A review of the theoretical and empirical literature, in: *Palgrave handbook of international trade*, pp. 497-531.

Roback, Jennifer (1982): Wages, rents, and the quality of life, *Journal of political Economy*, 1982, vol. 90.6, pp. 1257-1278.

Schulten, Thorsten, and Torsten Müller (2012): A new European interventionism? The impact of the new European economic governance on wages and collective bargaining, *Social developments in the European Union*, 2012, pp.181-213.

Seifert, Hartmut, and Heiko Massa-Wirth (2005): Pacts for employment and competitiveness in Germany, *Industrial Relations Journal*, vol. 36.3, pp. 217-240.

Sveikauskas, Leo (1975): The productivity of cities, 1975, *The Quarterly Journal of Economics*, vol. 89.3, pp. 393-413.

Tomassetti, Paolo (2017): From Fixed to Flexible? Wage Coordination and the Collective Bargaining System in Italy, *International Journal of Comparative Labour Law and Industrial Relations*, 2017, vol 33.4, pp. 527 – 551.

Van Hoorn, Andre, and Robbert, Maseland (2010): Cultural differences between East and West Germany after 1991: Communist values versus economic performance?, *Journal of Economic Behavior & Organization*, 2010, vol. 76.3, pp. 791-804.